

1 Objective-aligned evaluation inflates distributional gains 2 in single-cell drug retrieval

3 **Running title:** Objective–utility mismatch in single-cell drug retrieval

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Abstract

Single-cell drug retrieval is moving from mean signatures to distributional scores over treated populations, and routinely reports large gains under a metric of its own kind. We audit that practice under increasingly independent criteria, and show that mean-signature retrieval is the distributional family's zero-variance limit, not a competitor. Holding rankings, scorer and queries fixed and changing only the judge, distributional retrieval is decisively ahead under objective-aligned metrics, reverses under mechanism recovery, and beats its incumbent against an external functional oracle, where a scalar comparing no distributions reproduces most of it. Even blind external oracles reverse the winner by statistical form. We formalise when cell-level structure can pay off as two gates, differential response and recoverability; in patient tumours nobody constructed both open, yet no advantage appears: the mean already ranks most of what a subpopulation does. Distributional information here is real, and much smaller than objective-aligned evaluation makes it appear.

Introduction

Single-cell technologies have changed what it means to describe a drug's effect: it is not a single number but a distribution. Within a treated population, distinct subpopulations can respond in opposite directions, and the state that governs therapeutic outcome is often a minority one, a persister cell or a resistant clone. This premise drives the current wave of investment in virtual-cell models¹, single-cell foundation models²⁻⁴, and perturbation-response predictors⁵⁻⁷: if responses can be characterized at single-cell resolution, one should be able to rank candidate drugs better than methods that average the population into a mean signature, as the Connectivity Map does^{8,9}.

That premise is rapidly becoming method. Recent work has brought population-to-population distributional distances, optimal transport, maximum-mean-discrepancy, energy distance and the Wasserstein metric, into single-cell perturbation modeling, and these methods commonly report substantial gains over mean-based approaches. Yet they are almost always scored by distributional metrics of their own kind. When a distributional model is graded by a distributional yardstick, how much of the reported gain is a genuine improvement in decisions, and how much is the method and the yardstick sharing a mathematical target, has been asked too rarely¹⁰⁻¹², though the general hazard of steering by a proxy is not new¹³. We name this an objective-utility mismatch, and define it narrowly: *a retrieval system is evaluated by a criterion that measures the same statistical property its ranking maximised, so fidelity to the objective is read as evidence of downstream utility*. The term is scoped to retrieval evaluation and we do not extend it beyond that here. We are equally deliberate about how strong a word it licenses. Training on an objective and testing with a metric of the same family is not, by itself, a defect: a classifier fit by cross-entropy and scored by held-out log loss is honestly evaluated, because held-out data still tests generalization and the model can still lose. Retrieval is different, and sharper. There is no fitting step and no held-out set; the candidate is chosen by *maximizing a score*, and the evaluation then measures *that same score's* ground truth on *those same candidates*, so the ranking function and the evaluation function are one function evaluated once, with no estimation error to expose and no way to lose. We reserve the word circular for exactly this condition, and state it formally in Results.

Deciding whether “distributional methods are better” survives past the field’s own metrics requires an oracle that is both independent of the score and *semantically matched to the retrieval task*; getting the second requirement wrong is its own trap, since a similarity retriever graded on absolute potency is being asked a question it was never posed. The distinction we hold throughout is between objective fidelity and independent utility (Fig. 1). Objective-aligned metrics (Class A) legitimately test whether a method computes its own objective; task-proximal metrics such as mechanism-of-action recovery (Class B) are more independent but remain derived from the same upstream data; oracle-independent utility metrics (Class C), such as dose-response viability or longitudinal survival, are what a therapeutic claim ultimately requires, and are largely absent from the benchmarks these methods are evaluated on. An audit across six published method families, spanning signature retrieval^{8,9}, expression prediction^{5,6,14} and graph-based inverse design^{7,15,16}, finds objective-aligned evaluation to be the default in each. The central question is therefore not whether distribution-aware retrieval beats mean retrieval, but where objective fidelity ends and independent utility begins.

Making that boundary measurable calls for an instrument rather than a proposal: one family of population-to-population retrieval scores whose candidate rankings can be re-graded under criteria of increasing independence without being recomputed. We implement that family as EvalShift and release it with the code, and it is used here as a probe rather than offered as a method; the name is the finding, not an acronym, since what shifts across this study is the evaluation and not the rankings. Mean-signature retrieval, including Connectivity-Map cosine, is not a competitor but the zero-variance collapsed member of this same family: under the implemented kernel the two are numerically identical, with Hit@1 agreeing to three decimals. That unification is what makes the probe a controlled one, since only the judge then changes while the rankings, the scorer and the queries stay fixed, and we watch where the conclusion moves. Under objective-aligned metrics distributional retrieval looks decisively ahead; re-scored for mechanism recovery on the identical rankings the lead does not merely shrink but reverses; against an external functional oracle it does beat its incumbent, yet a scalar that compares no distributions reproduces almost the whole margin. The mismatch even reaches past metrics a method can see: two oracles built from the same proteins in the same cells, differing only in whether each is computed as a mean or as a distribution, hand victory to opposite methods.

What bounds the distributional gain is mechanistic, and it is the affirmative payload of this work. A predictor that adds a *single* delta vector to every cell has, by algebra rather than by experiment, two subpopulations with identical responses, so a distributional score has nothing to exploit; usefulness therefore requires that subpopulations both respond *differently* and can be *told apart* at all. We formalize this as a two-gate criterion, differential response and recoverability, and test it where it matters: in patient-derived glioblastoma tumours that nobody constructed, rather than in the cell-line mixtures we had assembled, which turned out to distort the biology in both directions at once and to flatter the premise both times. In real tissue both gates open, and the advantage still fails to appear, because the mean already ranks most of what the compartments do. The two gates are necessary, not sufficient, and what they omit is a third condition we name but do not measure as we measure them: the differential response must change *which candidate is preferred*, and in this tumour it largely does not.

Our angle differs from two adjacent lines of work. One provides distributional *prediction* methods, using optimal-transport or Schrödinger-bridge formulations to model perturbation responses^{17,18}; these

ask how to predict outcomes better, not how to rank candidate drugs, and do not test whether the evaluation is circular. A second compares distributional evaluation metrics for generative models^{19,20}, asking which metric better measures whether a generated distribution matches a real one; we ask the orthogonal question of how much of a reported gain is self-fulfilling when the retrieval score and the evaluation metric are the same object. To our knowledge neither this circularity nor the zero-variance identity has been stated before.

Three contributions follow, in the order the Results establish them. First, mean and distributional retrieval form one continuous spectrum of variance sensitivity rather than competing method classes, with mean-signature retrieval as its zero-variance endpoint; that is algebra, and it is what licenses holding the scorer fixed while the judge changes. Second, how large a distributional gain looks depends on how independent that judge is: decisive under an objective-aligned metric, reversed under mechanism recovery, and, for one pair of external oracles, settled by the oracle’s statistical form alone. Third, independent validation leaves a distributional contribution that is smaller than reported rather than absent, since against a semantically matched functional oracle the distributional score does beat its incumbent while a scalar comparing no distributions reproduces most of that margin. What this offers the field is a test for deciding whether a distributional gain is real or circular, a diagnosis of a failure mode that likely inflates results well beyond drug retrieval^{21,22}, and a roadmap of measurable conditions under which cell-level structure can pay off. The roadmap points somewhere specific, and it is the one encouraging thing we found: in a real tumour the subpopulation information a distributional score would need is *present*, and it is unsupervised clustering, not the data, that fails to recover it. That is an engineering problem, and engineering problems get solved. As single-cell foundation models proliferate, the question will only become more pressing.

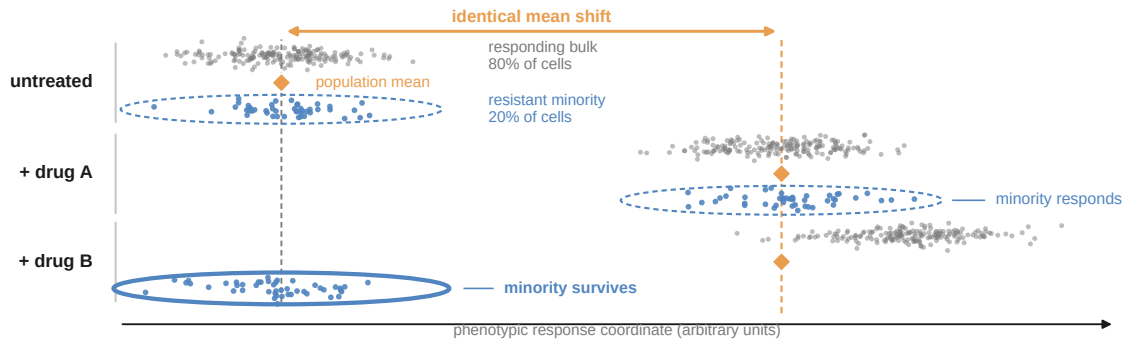
Results

Separating objective fidelity from independent utility requires an apparatus that varies the evaluator while holding everything else fixed. The apparatus is a single family of population-to-population retrieval scores, implemented as EvalShift, whose candidate rankings can be re-graded by criteria of increasing independence without recomputing the rankings themselves (Fig. 1). The same candidates, ranked once, are scored by the retrieval score’s own distributional objective (Class A), by a task-proximal biological criterion such as mechanism recovery (Class B), and by an oracle-independent functional read-out (Class C), so that from one analysis to the next the rankings, the scorer and the queries are held fixed and only the evaluator changes. Each apparent advantage thereby becomes a testable claim: a distributional gain that reflects better decisions should survive a more independent judge, whereas one that reflects only the scoring of a method with a metric of its own kind should not.

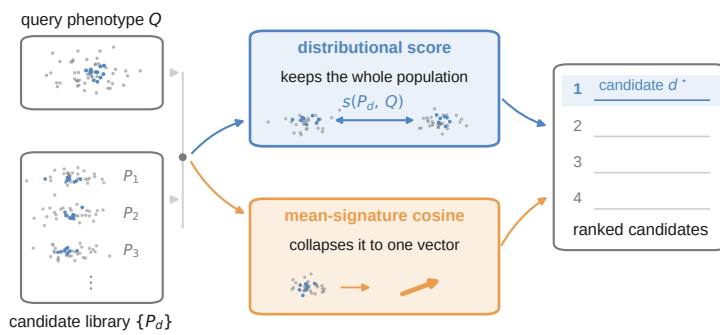
Mean and distributional retrieval are one continuum of variance sensitivity

We define the inverse retrieval task at the level of populations. For a target population Q and a library of candidate perturbation populations $\{P_d\}$, a retrieval score $s(P_d, Q)$ ranks candidates by similarity of effect. Mean-signature retrieval replaces each population with its mean differential-expression vector and scores the cosine (or correlation) between mean signatures, the CMap operation. Distributional retrieval instead scores the full populations with a distributional distance (energy, MMD, sliced-Wasserstein, or a subpopulation-coverage aggregate) and ranks by the negative distance.

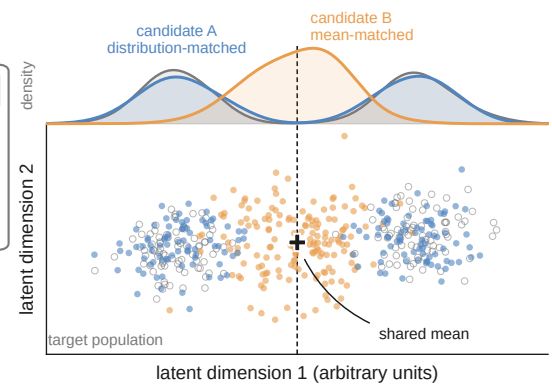
a Same mean shift, opposite fate for a hidden minority



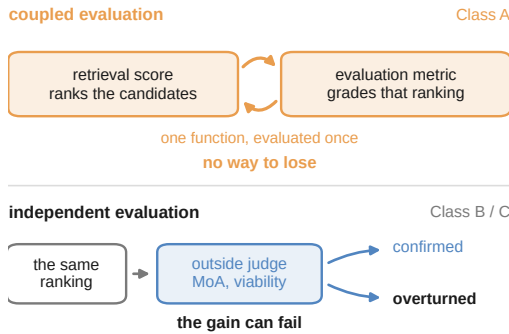
b One query, one library, two ways to score



c Means tie, distributions separate



d A score that grades itself cannot lose



e All three evidence classes, reported here

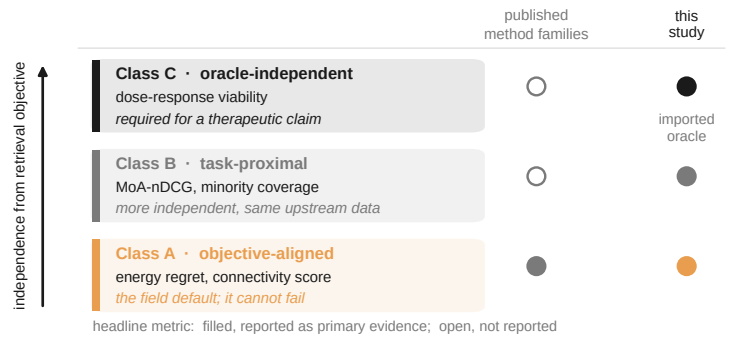


Figure 1: Distributional differences are visible, but their value depends on the evaluator. **a**, The premise, drawn on one response coordinate: two drugs produce an identical mean shift, but under drug B the resistant minority does not move and the responding bulk overshoots to compensate, so the population mean is preserved while the two drugs do opposite things to the cells that matter. Schematic. **b**, The inverse retrieval task and the two ways to score it: a distributional score keeps the whole candidate population, whereas mean-signature cosine collapses it to a single delta vector, and both return one ranked candidate list. **c**, Two candidate populations placed on the same sample mean, with the marginal densities above the scatter showing that one is bimodal and the other unimodal: means tie, distributions separate. **d**, When an evaluator is independent. A retrieval score graded by a metric of its own kind is one function evaluated once (coupled, Class A), so the ranking cannot lose; the same ranking sent to an outside judge can be confirmed or overturned (Class B/C). **e**, The three evidence classes ordered by independence from the retrieval objective, with the headline metric of published method families (filled for Class A only) against what this study reports (all three, its Class-C readout imported). Its Class-C readout is an imported, semi-real functional oracle built from dose-response profiles in cell lines the transcriptional data never saw; a Class-C claim measured on the same cells would require wet-lab validation.

These are not two unrelated method classes but two ends of one spectrum. Writing a population as its mean plus a zero-mean residual, the distributional distance reduces exactly to the mean-to-mean distance when the residual variance is taken to zero. Mean-signature retrieval is therefore the zero-variance collapsed member of the distributional family. Under the implemented cosine kernel this is not an approximation but an identity: mean-cosine retrieval and CMap cosine retrieval achieve identical Hit@1 (0.3885, to six figures 0.388492), because they compute the same operation, the cosine of mean-delta signatures. Distributional energy retrieval reaches 0.837 on the same queries. Both figures are unweighted macro-means over the seven (task $\times$ setting) cells of three retrieval tasks: controlled SciPlex3, cross-line SciPlex3 and Frangieh.

The collapse is one end of a continuous spectrum. The subpopulation-coverage aggregate carries a temperature β that interpolates between aggregation regimes, and on the controlled benchmark the interpolated distance rises monotonically from 0.7125 at $\beta \rightarrow 0$, exactly the mean-aggregated value, to 1.300 at $\beta \rightarrow \infty$, exactly the worst-case value (Fig. 2e). The global energy distance sits inside the same family, equalling the single-partition ($K = 1$) coverage aggregate exactly (Panobinostat, energy = coverage $_{K=1} = 0.1099$, difference 0). Energy, coverage and mean retrieval are therefore one axis of variance sensitivity rather than competing method classes, and mean-signature retrieval is its zero-variance endpoint (Fig. 2c,f).

This unification holds independently of any downstream utility metric: it is an algebraic fact about the scoring rules, not a claim about which method is therapeutically better. Two scope limits are explicit. First, the exact identity is specific to the cosine connectivity kernel; the canonical L1000 WTCS score is a rank-based (GSEA-style) enrichment of the same mean-collapsed signature, monotone in the same quantity but not algebraically identical to the cosine kernel implemented here (cmap_wtcs Hit@1 0.4635, which differs). Second, the identity concerns the scoring operation, not the choice of input population; the value of retaining the full distribution is exactly what the remaining results interrogate.

Objective-aligned evaluation inflates the gain

The collapse identity above has an empirical face. Across all retrieval metrics, mean cosine and Connectivity-Map cosine are perfectly correlated ($\rho = 1.00$), while the energy and coverage scores form a separate, near-orthogonal block (Extended Data Fig. 1, which also gives dataset scale and the distributional metric definitions). The correlation is computed over the 54,180 query–candidate scores that the 1,260 retrieval queries generate against their candidate libraries, at 43 candidates per query. The unit is the scored pair, not the query: an earlier draft called these “54,180 real queries”, which overstates the independent sample size by a factor of 43.

When candidate responses are *observed*, so that full candidate populations are available rather than predicted, distributional retrieval looks decisively stronger than mean retrieval. Energy-distance retrieval reaches Hit@1 0.837 against 0.388 for mean-cosine and the numerically identical CMap cosine; PCA-distance retrieval reaches 0.778 against PCA-mean’s 0.518 (Fig. 3). These are unweighted macro-means over the seven (task $\times$ setting) cells of the three retrieval tasks. Weighting cells by their query counts gives 0.887 against 0.421; we report the unweighted figure throughout, because it does not let the largest cell dominate.

Extending from observed populations to a partial-observed retrieval setting, we score candidates by an energy-based welfare regret, how much decision regret a ranking incurs against a latent welfare ob-

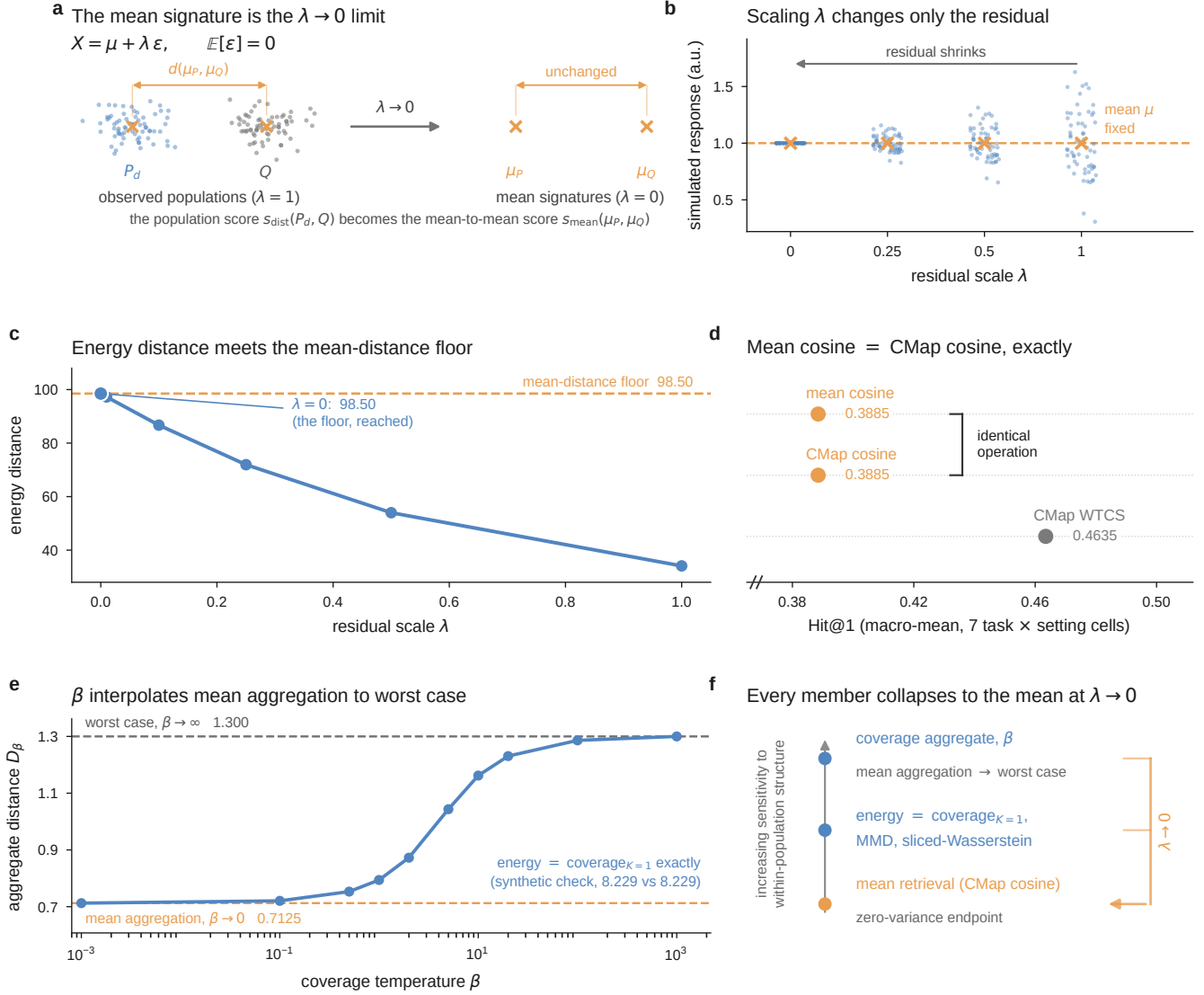


Figure 2: Mean and distributional retrieval form a continuous spectrum of variance sensitivity. **a**, Generative model $X = \mu + \lambda \epsilon$: the mean signature is the $\lambda \rightarrow 0$ limit. **b**, Scaling only residual variance leaves the mean fixed. **c**, Energy distance converges to the mean-distance floor as residual spread vanishes. **d**, Under the implemented cosine kernel, mean cosine and CMap cosine are numerically identical (0.388); WTCS is a rank-based sibling. **e**, The β -interpolation spectrum from mean-aggregation to worst-case, with energy equal to single-cluster coverage. **f**, Metric-family hierarchy: every limit collapses to the mean.

jective. Over all 765 partial-observed queries, the worst-case subpopulation-coverage scorer reduces median regret by **+0.118** against mean retrieval (mean +0.258), improving 72% of queries (paired Wilcoxon $p = 1.6 \times 10^{-66}$). We report the full query set here rather than the gate-recommended subset of it: the two are indistinguishable (+0.119 on the 621 recommended queries), and selecting a headline on a gate-selected subset would be exactly the analytic degree of freedom this paper audits.

By the standard of these metrics, distributional retrieval is far ahead. But the metric here is objective-aligned: the energy welfare regret and the energy retrieval score measure the same distributional object. This result establishes objective fidelity: distributional retrieval computes a discriminative objective, and observed heterogeneous populations contain real distributional signal. It is a *distributional retrieval signal under an objective-aligned proxy*, not evidence of independent therapeutic utility. Whether that signal survives a more independent judge is the question the next sections answer.

Task-proximal metrics collapse the apparent gains

This is the central result (Fig. 4). Both sides of the comparison use one scorer on one query set, so the only thing that changes is the class of metric doing the judging. Selecting the best scorer separately for each metric class, on different query subsets, is an analytic degree of freedom this task readily permits, and it would report +0.119 against -0.013: it *understates* the collapse. We fix both and report the larger number.

On the 600 partial-observed queries where both metric classes are defined (MoA recovery is undefined for the remaining 165; Methods), the worst-case coverage scorer achieves a Class-A regret reduction of +0.129 (median; mean +0.275). Re-scoring the *identical* rankings under mechanism-of-action recovery nDCG, the same two statistics are 0.000 (median) and -0.037 (mean; Fig. 4a). We give both, because they say different things and both are load-bearing: on the median query the advantage does not shrink but vanishes *exactly*, and averaged over queries it turns *negative*. The reversal is not an artefact of a heavy tail: 41.3% of queries *lose* mechanism recovery under the distributional score against 34.2% that gain it (Wilcoxon $p = 2.4 \times 10^{-4}$). The distributional score is not merely no better; it is worse. All five distributional variants behave the same way (Class-A gain +0.056 to +0.129, Class-B loss -0.011 to -0.037), so the conclusion does not depend on which variant is chosen.

Projecting the real datasets onto the retrieval task, the distributional advantage in minority-state coverage across all 239 real-data tasks is statistically real and practically negligible: mean +0.0018, Wilcoxon $p = 1.3 \times 10^{-7}$ on the 137 tasks with a nonzero difference, on a metric whose observed range is [0.60, 1.00]. In 102 of the 239 tasks the difference is *exactly* zero, because both method families select the same drug. And the effect is confined to the mixtures we constructed: the mean gain is +0.0043 on the 90 constructed cross-line tasks, against +0.0004 within-line ($n = 90$), +0.0004 on Frangieh ($n = 23$), +0.0003 on CD34+ ($n = 12$) and +0.00003 under predicted candidates ($n = 24$) (Fig. 4g).

An independent divergence audit reaches the same verdict and numerically the same effect: a coverage gain that is real, monotone in divergence, and around +0.002, which is to say too small to act on. Two analyses that share no code path converge on it.

A dominance count on this data counts noise excursions. The conventional summary would count tasks whose gain clears a threshold, here +0.01. That count is not interpretable (Methods): the threshold sits inside the noise band of the difference distribution and is one-sided, and the seed is not a replicate, since it re-draws which drugs are held out and re-clusters the minority subpopulation. Holding the drug

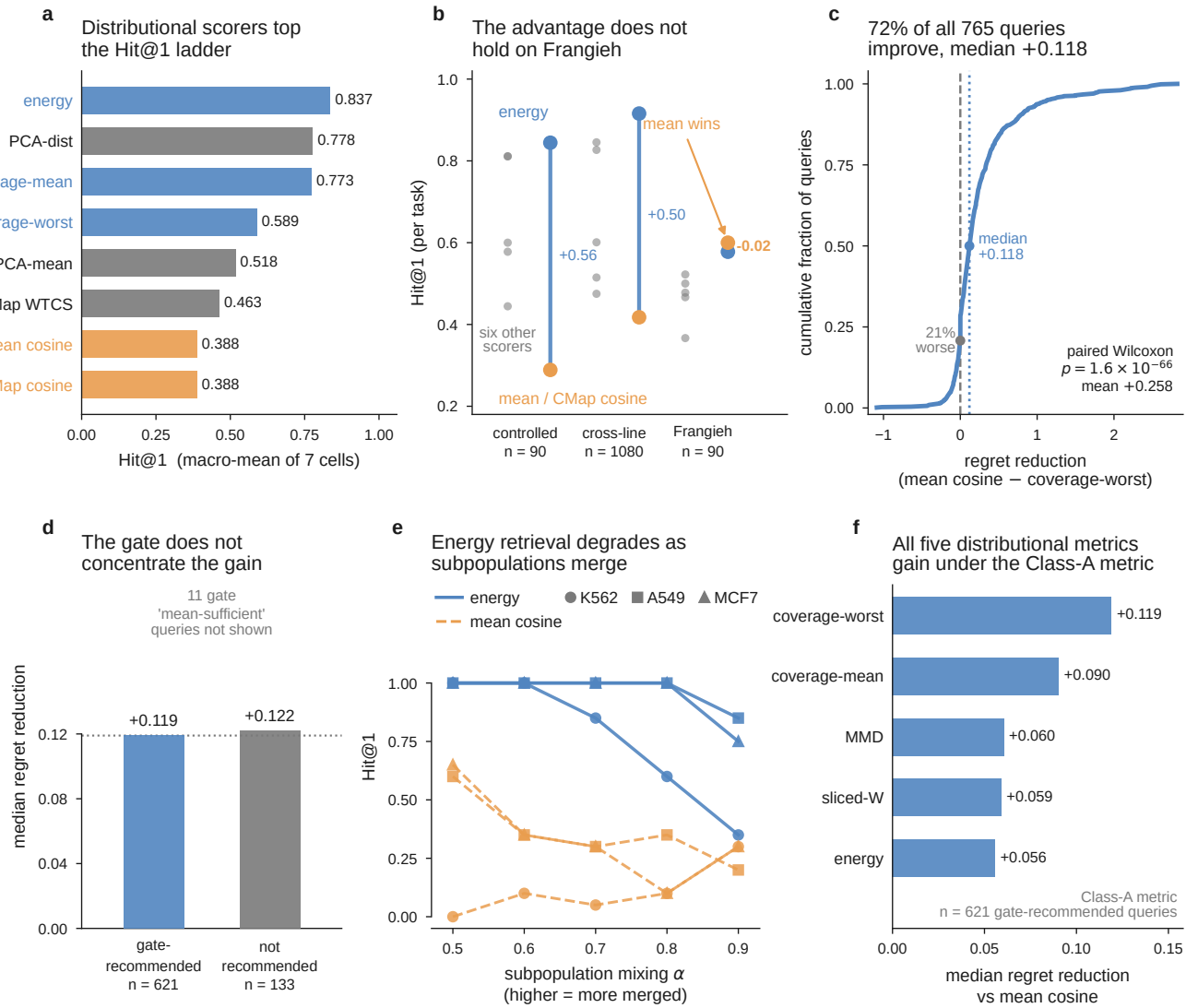


Figure 3: Under objective-aligned metrics, distributional retrieval looks decisively stronger. **a**, Hit@1 ladder: energy 0.837 down to mean/CMap cosine 0.388, unweighted macro-mean over the seven (task $\times$ setting) cells of three retrieval tasks. **b**, Per-task Hit@1. The advantage holds on the controlled and cross-line SciPlex3 tasks but *not* on Frangieh, the only natural real dataset in the macro-mean, where mean/CMap cosine (0.600) beats energy (0.578). The macro-mean hides this counterexample; we state it here and in the text. **c**, Per-query regret reduction over *all* 765 partial-observed queries: median +0.118, 72% improved (the +0.119 quoted elsewhere is the 621-query gate-recommended subset, and the two are indistinguishable, which is panel **d**'s point). **d**, Recommended (+0.119) versus non-recommended (+0.122) subsets: the gate does not concentrate the gain. **e**, Alpha-crossover. The energy advantage does not close monotonically as the two subpopulations merge, and it does so in only one of the three cell lines (K562, from 1.00 to 0.05 in Hit@1 as α rises from 0.5 to 0.9); in A549 and MCF7 the gap is as large or larger at high α . We report the per-line curves rather than a pooled trend for that reason. **f**, All five distributional metrics show positive Class-A regret reduction.

fixed and resampling the seed ten times, only **2 of the 10** threshold-crossing cross-line tasks stay above it in at least 8 of 10 seeds, while control drugs that clear it at *no* recorded seed clear it in 1 to 4 of 10 resamples. We report the distribution instead. The two survivors are a hypothesis rather than a result: A549→MCF7 Abexinostat ($+0.090 \pm 0.013$) and Belinostat ($+0.081 \pm 0.027$), both HDAC inhibitors, both above threshold in 10 of 10 seeds. With $n = 2$, one cell-line pair and one drug class, this is the only place in this study where a distributional advantage on real data survives being checked, and it is suggestive and no more.

Crucially, a diagnostic gate designed to *concentrate* the advantage onto the queries where distributional information should help does not do so. The median energy-proxy regret reduction is $+0.119$ on the gate-recommended subset ($n = 621$) against $+0.122$ on the non-recommended subset ($n = 133$; the remaining 11 of the 765 queries are labelled mean-sufficient). The gate does not separate the cases where the distributional score helps from those where it does not, even on the proxy metric it was tuned against, and if anything it selects slightly against them.

The simplest demonstration is at the dataset level. The divergence criterion holds that distributional retrieval can help only when subpopulation responses diverge, operationalized as a response cosine below ≈ 0.9 . Measuring where each dataset actually sits on that axis, cross-line SciPlex3 lies at cosine 0.032, Frangieh at 0.709, and CD34+ at 0.186. All three clear the gate. Yet the distributional score wins only on cross-line: on Frangieh, the only natural real dataset, the mean-signature incumbent wins (Hit@1 0.600 versus 0.578), and CD34+ is our *negative control*, where the mean wins decisively. A criterion that admits the positive anchor and the negative control alike is not a criterion, and the dataset-level divergence rule therefore fails on its own terms.

A direct audit makes the failure concrete (Fig. 4e,f). The gate’s structure-reliability axis is anti-correlated with true response divergence (Spearman $\rho = -0.21$, $p = 3.8 \times 10^{-9}$), moving opposite to the quantity it is meant to track; its preference-conflict axis is only weakly aligned ($\rho = +0.18$); and the recommendation label does not separate high- from low-divergence queries (Mann–Whitney $p = 0.090$). Bypassing the gate and stratifying directly by true divergence, the highest-divergence stratum shows a statistically significant but negligible minority-coverage gain (median gap rising from $+0.0009$ in the lowest quartile to $+0.0018$ in the highest, all Benjamini–Hochberg $q \approx 0$, against a metric operating range of $[0.89, 0.99]$: about 0.13 standard deviations). Mechanism recovery is null in the highest stratum ($q = 0.57$, $n = 143$) and *significantly negative* in the lowest (median gap -0.030 , $q = 5.8 \times 10^{-5}$, $n = 166$): where responses are least divergent, the distributional score actively degrades annotation recovery. The deficit is in the data and the task, not only in the gate.

The obvious objection is that a null is merely an underpowered study, and the two metric classes are indeed unequally powered (Extended Data Fig. 2): the coverage metric is fully powered (achieved power 1.00), while MoA recovery is not (0.06; 80% power would need roughly 20,800 queries against the 143 available). We therefore do *not* claim to have proven the absence of a small MoA effect. We claim something weaker and sufficient: where the theory predicts the largest gain there is no detectable gain, and where responses are least divergent the effect that *is* detectable runs against the distributional score.

The conclusion is deliberately strong and deliberately bounded: apparent distributional gains are metric-dependent and cannot be interpreted as therapeutic utility without independent validation. One scorer, one query set and one set of rankings is “far ahead” or “worse than the incumbent” depending

only on which metric class judges it. That dependence, not a verdict on any single method, is the finding.

A functional oracle shows the gain is mostly magnitude

Class B is task-proximal, not independent, so the ladder is incomplete without an external functional readout. Choosing that readout is where a Class-C analysis is most easily got wrong, and the wrong choice is the tempting one.

The oracle must match the task. A retrieval score is asked which candidate’s response distribution most resembles the query’s. It is *not* asked which candidate is most cytotoxic. Grading a similarity retriever on absolute potency therefore tests a question it was never posed: handed a weak query, a correctly working retriever *should* return other weak drugs. We report that mismatched comparison only as a confounder audit (below), because the conclusion it invites, that retrieval is anti-aligned with therapeutic value, does not follow from it.

The oracle that does match the task is drug–drug functional similarity: do the query and the candidate *behave* alike in an independent assay? GDSC2^{23,24} supplies this directly. Each compound has a dose-response AUC profile across roughly a thousand cell lines, and two drugs with similar cross-cell-line sensitivity patterns are functionally similar however potent either happens to be. We define $\text{oracle}(q, c) = \text{Spearman}(\text{AUC}_\bullet(q), \text{AUC}_\bullet(c))$ over the 966 GDSC2 cell lines that remain after *excluding* the three SciPlex3 lines, so the oracle is never evaluated in the context the transcriptional query came from. Because a correlation is centered per drug, a compound’s overall potency level cannot drive it; only its selectivity *pattern* can. Cell lines differ enormously in general drug sensitivity, and left uncorrected that shared main effect makes almost every drug pair look alike (85% of drug pairs positive, median +0.178, versus 52% and +0.015 once corrected); we therefore subtract each cell line’s mean AUC, estimated across all 286 GDSC2 compounds rather than only the matched ones, before correlating.

That correction decides the result, and we report the dependence rather than bury it. Without it, energy falls to +0.097 and drops *below* the mean-signature incumbent’s +0.138, reversing the one affirmative finding of this section; the magnitude scalar (+0.217) beats both. Both oracles are given in full in Supplementary Table 4. The correction is defensible on its face, it was specified on the reasoning above before the retrieval results were inspected, and leaving a shared cell-line main effect in an oracle would be an error of its own. But it is an analytic choice, and a paper arguing that the choice of criterion decides the winner cannot exempt its own criterion from that finding. The honest statement is that the Class-C positive below holds under the oracle we argue is correct and not under the oracle we argue is wrong.

On this oracle, distributional retrieval beats its incumbent (Fig. 4h). Across 103 leave-one-drug-out queries, energy retrieval attains $\rho = +0.276$, against +0.083 for the correctly specified mean-signature incumbent: a threefold margin, in the same direction in all three cell lines. This is the first point in this study at which the distributional score beats its incumbent on a criterion no scorer can see, and we report it as such.

It is nevertheless not a distributional result. Rank the candidates instead by how closely their response *magnitude* matches the query’s, a query-dependent scalar that compares no distributions at all, and it reaches $\rho = +0.232$. Energy’s edge over that scalar is 62 of 103 queries, which even under an anticonservative paired test is not significant ($p = 0.073$). Partial out the magnitude channel and energy retains +0.097 (per line +0.131, +0.053, +0.162: positive in all three, but small).

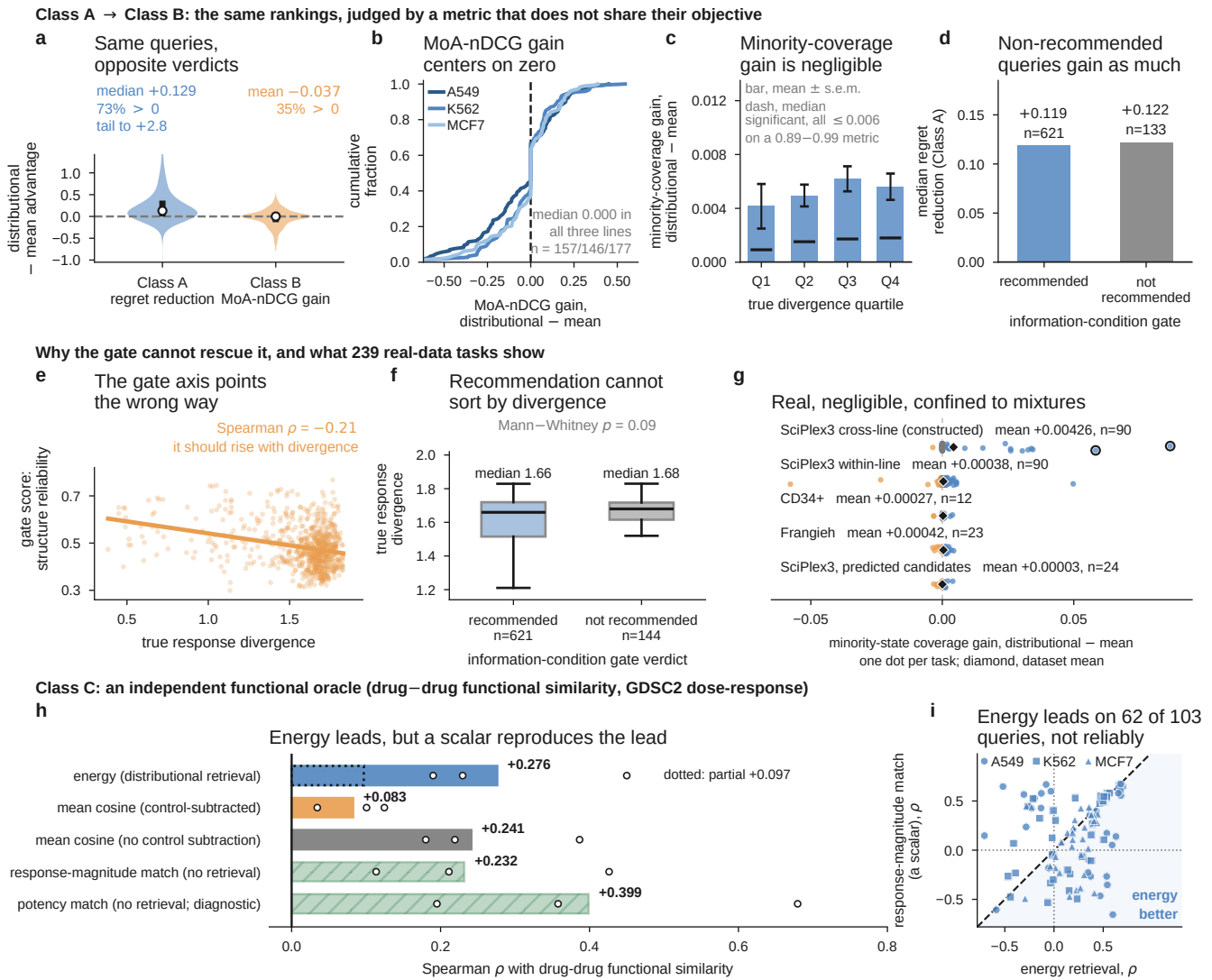


Figure 4: Independent evaluation collapses the apparent advantage, and what survives it is mostly magnitude. The metric-class ladder, read row by row. **Row 1, Class A → Class B.** **a**, One scorer, one query set, only the class of metric changes: a Class-A regret reduction of +0.129 becomes a Class-B MoA-nDCG gain of -0.037. **b**, MoA-nDCG gain by cell line, uniformly at or below zero. **c**, Minority-coverage gain by divergence quartile: significant (all Benjamini–Hochberg $q \approx 0$) and negligible, every bar ≤ 0.006 on a metric whose values run 0.89 to 0.99. Bars are quartile means with s.e.m.; dashes mark the medians the text quotes. **d**, The gate does not concentrate the advantage (+0.119 on 621 recommended queries versus +0.122 on 133 non-recommended). **Row 2, why the gate cannot rescue it.** **e**, The gate’s structure-reliability axis is *anti*-correlated with true response divergence ($\rho = -0.21$), moving opposite to the quantity it is meant to track. **f**, Recommended and non-recommended queries have indistinguishable divergence (Mann–Whitney $p = 0.09$). **g**, The real-data coverage advantage, one dot per task across all 239 tasks: real but negligible (overall mean +0.0018) and confined to the constructed cell-line mixtures (+0.0043 against +0.0003 in the natural data and +0.00003 under predicted candidates). We plot the difference rather than a dominance count; only 2 of the 10 threshold-crossing cross-line tasks survive seed resampling (circled). **Row 3, Class C, an independent functional oracle.** 103 leave-one-drug-out queries, SciPlex3 at 10 μ M. The oracle is *drug–drug functional similarity* measured out of context, not absolute potency (Methods). **h**, Energy reaches $\rho = +0.276$ against +0.083 for the correctly specified mean incumbent, the first oracle-invisible criterion here on which the distributional score wins. It is not a *distributional* win: a query-dependent scalar matching only response *size* reaches +0.232 (hatched), and partialling that channel out leaves energy with +0.097 (dotted). The longest bar, potency matching at +0.399, is a diagnostic rather than a recommender, since it needs the query drug’s own measured potency. Open circles, per-cell-line medians. **i**, Per query, energy against that scalar: energy wins 62 of 103 queries, and the nominal $p = 0.073$ is anticonservative because queries within a line share a candidate pool. The mismatched absolute-potency comparison is reported only as a confounder audit (Extended Data Fig. 5).

Even a *query-independent* ordering carries comparable alignment: ranking candidates by their own response magnitude, an ordering identical for every query that performs no retrieval whatsoever, gives $\rho = -0.330$, so the same fixed ordering reversed gives $+0.330$. We report its magnitude rather than calling it a better recommender, and deliberately so: its *sign* would have to be known in advance, and a paired comparison in fact favours energy on 65 of 103 queries even though its marginal median is the larger of the two. The point is only that a constant ordering is not far behind a distributional score on this oracle. Finally, a single scalar matching the candidate’s potency to the query’s reaches $+0.399$, above the distributional score, though it is a diagnostic rather than a usable recommender because it requires the query drug’s own measured potency; what it establishes is that the functional oracle retains a substantial potency-level axis, which is the channel the mismatched comparison below confounds entirely.

The picture is therefore layered rather than negative. Distributional retrieval carries genuine functional signal that the mean signature does not; almost all of that signal is a response-magnitude channel that a scalar reproduces; and the residual attributable to the *distribution* is roughly $+0.097$, about a third of energy’s raw $+0.276$ and positive but small. We deliberately do not express this as a ratio against the Class-A gain: a Spearman correlation against an external oracle and a Hit@1 or regret improvement against an objective-aligned proxy are different quantities on different scales, and dividing one by the other would manufacture a number rather than measure one. The comparison that is legitimate is within the oracle, and there two thirds of the distributional score’s advantage is magnitude. Objective fidelity does not make the distributional gain imaginary. It makes it look far larger than it is.

One operational lesson is specific to baselines, and the other is deferred to the Discussion, where we state the controls this study recommends. A mean-signature baseline scored *without* control subtraction reaches $+0.241$, close to energy and far above the correctly specified incumbent’s $+0.083$, because it measures baseline cell identity rather than drug response. A study using it as its incumbent compares against a broken baseline and will overstate its own margin.

Confounder audit: why the mismatched oracle inverts. Scored against *absolute* potency, energy retrieval is anti-correlated ($\rho = -0.520$, same sign in all three lines) while the query-independent magnitude scalar reaches $+0.692$. This is arithmetic, not pathology. Response magnitude predicts potency on its own, and the energy *distance* between query and candidate largely tracks the candidate’s own magnitude ($\rho = +0.791$), because big-response populations sit far from everything, so ranking nearest-first returns the weakest candidates. The same magnitude channel drives both oracles; this is not evidence that retrieval opposes therapeutic value, and we do not claim that it is.

Scope. This oracle is semi-real: viability is measured in a separate bulk assay, joined by compound identity rather than shared cells, under a different dose regime, so it constrains what a transcriptional score recovers about drug function without being a clinical endpoint. The 103 queries within a cell line share a candidate pool, so paired tests across them are anticonservative; the honest unit is the three cell lines, and we give per-line values throughout. The strongest limit is the one stated above, that the affirmative result is not robust to the cell-line centering: under the uncorrected oracle energy reaches only $+0.097$, below both the incumbent ($+0.138$) and the magnitude scalar ($+0.217$). What survives both constructions is the negative, that under neither oracle does the distributional score beat the scalar that compares no distributions.

Oracle shape not biology decides which method wins

The Class-C result above is not the only external readout available, and the second one disagrees with it. Frangieh Perturb-CITE-seq measures 20 surface proteins in the *same* 218,331 cells whose RNA the scorers read, giving an oracle that is independent of the transcriptome, invisible to every scorer, and free of the cross-assay join the GDSC oracle needs. Scored against it, the *mean* beats energy (+0.242 against +0.146), in all three immune conditions. Taken at face value, one external oracle says distributional retrieval wins and another says it loses.

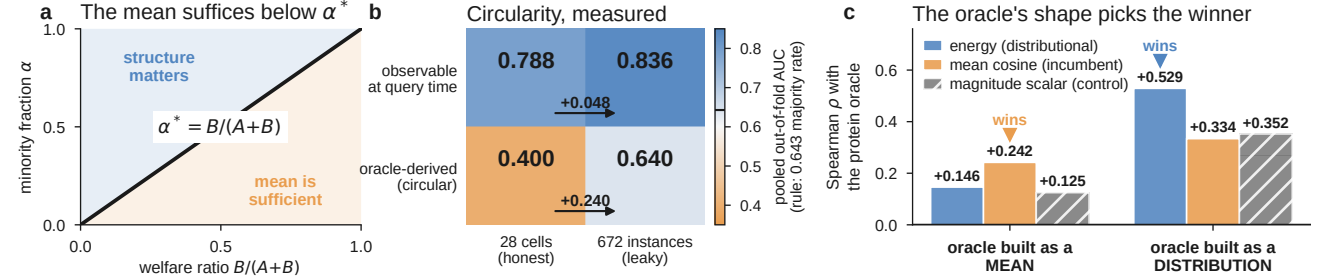
They are not in conflict. Look at what the protein oracle is: cos of the two knockouts' *mean* protein deltas. It collapses the protein readout to a mean, and the mean-cosine RNA scorer computes that same statistic in the other modality. A mean-shaped oracle is structurally matched to a mean-shaped scorer, and energy is being graded on a criterion that discarded the very thing it compares. That is the mirror image of the Class-A alignment this paper is about, arising now between two *external* criteria, and it is testable directly.

We therefore rebuild the oracle from the same cells and the same proteins, twice, changing only its statistical form (Fig. 5c; Methods). One version is the mean cosine above; the other is the energy distance between the two knockouts' protein-response *cell clouds*, both referred to that condition's control cells so it measures response and not baseline identity. The two RNA rankings under test are byte-identical across the two columns. Nothing about the scorers, the cells, the proteins or the queries changes. Only the shape of the judge changes.

The winner swaps. Under the mean-shaped oracle the incumbent leads (+0.242 against +0.146); under the distribution-shaped oracle the distributional score leads, and by more (+0.529 against +0.334). The reversal holds independently in each of the three immune conditions (Control, IFN γ and Co-culture; $n = 213, 227$ and 219 queries). No scorer can see either oracle, so this is not circularity in the Class-A sense of a method grading its own objective. It is something we did not anticipate and consider the sharpest result in this paper: even when the criterion is external, independent and blind, its own statistical form selects the winner. A field that validates distributional methods against distributional readouts, and mean-based methods against mean readouts, will find each vindicated, and neither finding will be about biology.

The magnitude control survives this, and refines it. An energy distance tracks a candidate's own response magnitude ($\rho = +0.791$), and both distributional objects here are energy distances, so a magnitude-to-magnitude channel could in principle produce the swap with no distribution ever being compared. It does not, quite. The magnitude scalar does climb steeply when the oracle turns distributional (from +0.125 to +0.352), which is the confound behaving exactly as predicted and is itself worth reporting. But energy still leads that scalar by +0.177 under the distributional oracle, against only +0.022 under the mean-shaped one. So there is a genuine distributional signal in these cells, and *it is visible only to a criterion that is itself distributional*. Whether that makes it a real advantage or an unfalsifiable one is the question this paper leaves open, and we prefer to leave it open than to answer it with a criterion of our own choosing.

From the inside: analytically, inside a synthetic benchmark, and between two real oracles



From the outside: our constructed benchmark, checked against tissue nobody assembled

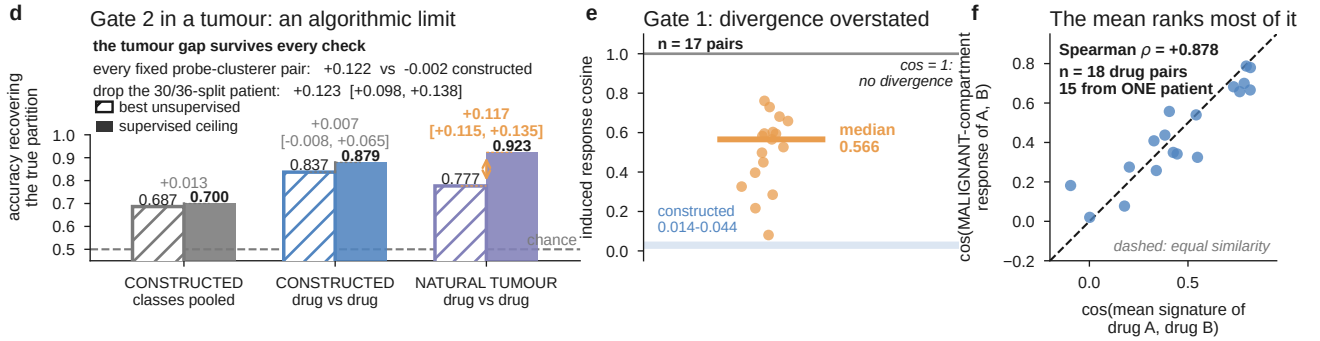


Figure 5: The statistical form of the criterion, not the biology, decides which method wins; and in real tissue both gates open without an advantage following. Row 1. **a**, HIR-Bench's analytic flip boundary $\alpha^* = B/(A+B)$ separates the regime in which the mean is a sufficient statistic from the one in which subpopulation structure can change the decision. **b**, Circularity measured rather than argued: features a method can observe at query time reach AUC 0.788 against a 0.643 majority-class rate, whereas features derived from the benchmark's own latent utility matrix, of which the label is also a function, are at chance (0.400). Pseudo-replication inflates the circular set by +0.240 and the honest one by only +0.048. Effective n is 28 parameter cells, not 13,440 (Methods). **c**, The same effect between two external criteria built from the same proteins in the same cells, differing only in statistical form. **The winner swaps**: the mean incumbent leads under the mean-shaped oracle (+0.242 against +0.146), the distributional score under the distribution-shaped one (+0.529 against +0.334), in each of the three immune conditions. Hatched, the magnitude-matching control, which compares no distributions: it climbs from +0.125 to +0.352 as the oracle turns distributional, yet energy still leads it by +0.177 there against +0.022 under the mean-shaped oracle. Row 2, the two gates on ZhaoSims2021, ten glioblastoma patients (Methods). **d**, Gate 2 posed as the same drug-response question in both settings (36 splits, 4 patients). Pooling drugs into classes holds the constructed ceiling at 0.700 (grey); one drug against one drug, the same cells give 0.879. All three arms are measured at the same granularity and under the same representation, so they are like for like; the 0.692 quoted for this construct elsewhere is the earlier in-fold run (Fig. 6e). In a real tumour the information is present (ceiling 0.923) and unsupervised clustering does not reach it (0.777; median paired gap +0.117, 95% CI [+0.115, +0.135] by patient-level bootstrap, against +0.007 constructed; +0.122 against -0.002 with no method selection anywhere; +0.123 [+0.098, +0.138] without the dominant patient). **Gate 2 in real tissue is an algorithmic bottleneck, not an informational one**. 30 of the 36 splits come from one patient and 39.9% of cells fail the assignment margin, so both numbers are optimistic. **e**, Gate 1: induced response cosine, median 0.566 over 17 patient-drug pairs, against 0.014–0.044 in our constructed mixtures. **f**, The premise: over 18 within-patient drug pairs, mean-signature similarity ranks malignant-compartment response similarity at Spearman $\rho = +0.878$, and the disjoint form of the same question, ranking the malignant compartment by the *myeloid* compartment rather than by a mean that contains 43% of the malignant cells, gives +0.835. No p -value is given, since 15 of the 18 pairs come from one patient. Both gates open here and the distributional advantage still does not appear: the two gates are necessary, not sufficient, and the condition that fails is the third one, decision relevance (Results).

379 The information condition is a two-gate criterion

380 The preceding results pose a mechanistic question: why is the distributional signal large on observed
381 populations yet absent under independent metrics? The answer is a *candidate-response information con-*
382 *dition* with two distinct gates. A distributional score can only exploit subpopulation structure whose
383 subpopulations (i) actually respond *differently* in the candidate populations supplied to the retriever,
384 and (ii) can be told apart from those populations at all. Both gates are closed *in the constructed cell-line*
385 *data this section uses*, which is why the observed-population result is an upper bound rather than an op-
386 erating point there. That qualifier is load-bearing and we did not originally have it: on natural patient
387 tumours both gates turn out to be open, and the last Results section is the one that corrected us.

388 **A third condition, which we name here rather than measure as we measure the first two.** Gates
389 1 and 2 concern whether the information a distributional score needs is present and reachable. Neither
390 asks whether it *matters*. We therefore name a **Gate 3, decision relevance**: the differential response must
391 change *which candidate is preferred* under the outcome the decision is judged on, over and above what the
392 mean signature already ranks. Its status in this paper is different from the other two and we mark that
393 difference wherever it appears. Gates 1 and 2 are measured directly, on labelled mixtures and on patient
394 tumours. Gate 3 is inferred, from three lines of evidence that were collected to answer other questions:
395 in a real tumour the response of one compartment already ranks the response of another at Spearman
396 $\rho = +0.835$, a value dominated by one patient (Fig. 5f); against the independent functional oracle a
397 scalar comparing no distributions reproduces most of the distributional score’s margin (Fig. 4h,i); and
398 no true-divergence stratum shows a positive mechanism-recovery gain (Fig. 6g). We propose Gate 3 as
399 the condition that explains why opening the first two is not enough, and we do not claim to have tested
400 it as such.

401 **Gate 1, differential response.** We test what generated candidate populations actually contain, using
402 predict-then-rank experiments in which candidate responses are produced by a perturbation predictor
403 rather than observed (Fig. 6). Across three predictor families the distributional advantage over mean
404 retrieval is small and *not consistent in sign*: the nDCG@10 delta is -0.016 (average-effect), -0.071 (nearest-
405 neighbour) and $+0.027$ (CPA-style linear-latent, a surrogate for the latent-arithmetic family; we do not
406 benchmark a trained scGen). The distributional score also wins on Hit@1 for two of the three. We
407 therefore do not claim the advantage is uniformly absent under prediction, only that it is small, unstable,
408 and does not survive the independent metrics of the previous sections.

409 The structural explanation is not the intuitive one. Predicted candidate populations do *not* collapse
410 subpopulation structure. Synthesizing a predicted population by applying the predicted effect to the
411 query context’s real control cells, which is what the latent-arithmetic models in this class actually do,
412 the subpopulation-variance ratio of predicted populations is 0.142 against 0.138 for the real blended
413 candidate population that a scorer ranks; the latent model even sharpens it, to 0.462 (Fig. 6c). Structure
414 survives.

415 What does not survive is *divergence*, though the way we first measured it was doubly wrong and
416 the corrected version says something sharper. Measuring the induced response cosine between the two
417 subpopulations of a candidate population, $\cos(d_{\text{maj}}, d_{\text{min}})$ with each subpopulation referred to its own
418 matched control, real candidate populations respond near-orthogonally (**0.014**) while predicted ones re-
419 spond in largely aligned directions (0.186 average-effect, 0.239 linear-latent, 0.366 nearest-neighbour).

Those predicted values do not mean what we took them to mean, for two reasons given below: the populations are cross-context blends, and for a latent model the quantity is contaminated by the generator’s reconstruction error. We retain them because the structure-preservation conclusion above does not depend on them, and we do not build any claim on them.

The limit of this behaviour is algebraic, not empirical, and the algebra is what carries the argument. Call a predictor additive if it adds a *single* delta vector to *every* cell of the query context, whether in gene space or through an invertible latent map. Its two subpopulations then have *identical* response deltas, $d_{\text{maj}} = d_{\text{min}} = \delta$, so their induced cosine is exactly 1 and the divergence a distributional score could exploit is exactly zero by construction. No amount of training changes this. Additivity is thus a *sufficient* condition for Gate 1 to be shut, and it is a property of a model class, not a claim about any particular software.

Which deployed predictors fall in that class is a separate, empirical question, and the within-context measurement can now answer part of it. Every number below is the model’s *true* induced divergence, taken against its own baseline, and every model is first required to pass a learning check: the cosine between its predicted mean delta and the true one. A model that has not learned the drug effect cannot be said to open or shut any gate, and its divergence is noise structure. We enforce that rather than assume it.

Real treated cells, split by the same control-derived cell state, give the target: their two subpopulations respond at a cosine of 0.205, so within-context subpopulations in real data do respond very differently. Against that, the average-effect predictor (learning check 0.502) and the linear-latent model (0.343) both return 1.000000 exactly, reproducing none of it, as the algebra requires. The published scGen variational autoencoder (learning check 0.420), which this paper previously declined to say anything about, returns 0.972: its nonlinear decoder buys a small departure from strict additivity, and it reproduces roughly 3% of the divergence real cells show. *A deployed latent-arithmetic model is therefore near-additive in practice, not only in principle*, and Gate 1 is essentially shut for it.

A genuinely non-additive predictor does open Gate 1, but only a crack. A predictor whose displacement depends on the cell it is applied to should, if the two-gate account is right, show a divergence below the additive ceiling of 1. We tested an entropic optimal-transport map, the non-parametric core of the neural-OT perturbation models^{17,18}, run in a latent space as every deployed neural-OT model is. It passes its learning check (0.352, above the 0.30 bar) and its *true* induced divergence is 0.900 ± 0.101 , significantly below the additive ceiling, so a non-additive predictor genuinely gives different subpopulations different responses. But real treated cells sit at 0.205, so the map closes only about an eighth of the distance from the ceiling to what biology shows, and its learning is weak enough that part even of that crack is prediction noise.

Two deployed models meet the same wall from opposite sides. The published CPA reconstructs cells acceptably ($R^2 \approx 0.39$) but does not learn the drug-specific effect (predicted-versus-true mean-delta correlation 0.08), so it sits below the learning bar and its divergence is not reportable; neither five times more cells per drug nor conditioning its perturbation embedding on 2048-bit Morgan fingerprints in place of a learned vector rescues it (learning 0.08 and 0.03), which locates the failure in the adversarial objective rather than the drug representation. CellFlow²⁵, an optimal-transport flow-matching model conditioned on the same fingerprints, does learn within context (mean-delta cosine 0.24 over twenty drugs, 0.81 for

the HDAC inhibitor AR-42) and opens Gate 1 (induced divergence 0.85). But the clean test requires prediction into a held-out context, and there its learning collapses to 0.03 at a matched budget. That collapse is not an artefact of a mean-based check: re-scored with the gene-resolved and distributional metrics used by community virtual-cell evaluations²⁶, the held-out-context predictions match the true differential-expression direction only at chance (0.52 against 0.70 within context), recover 7% of the true differential genes, and sit farther from the true treated population than the unperturbed control does (energy ratio 2.1 against 0.98). It is instead mechanistic: per drug, leave-one-context-out learning tracks that drug’s cross-context response similarity (Pearson $r = 0.85$), so a predictor conditioned on chemistry alone recovers only the context-shared part of a response, which on this dataset is small (median cross-context cosine 0.04) and near zero for exactly the response-divergent drugs the retrieval task selects.

Whether opening Gate 1 buys retrieval gain therefore stays open, for an instructive reason. Re-running the cross-line predict-then-rank with the OT map as the candidate-response source, its predictions are too poor to retrieve with at all: Hit@1 is 0.05 under *every* rule, mean cosine included, against 0.88 for the additive average-effect predictor. **We lack a predictor that both predicts accurately across contexts and opens Gate 1**, which is exactly what a clean test of the second half of the two-gate account needs. That we could not build one is consistent with the rest of this paper, in which the accurate part of every predictor is its mean and the part that departs from the mean is noise, but it is a limitation rather than a falsification and we present it as such.

One measured value is an artefact of the generator, not of the biology. An additive predictor is pinned by the algebra to return exactly 1.000000, and asserting that value is what exposed two errors in our own first design: the candidate populations were cross-context blends, and a latent model’s response must be referred to the right baseline (Methods). The second of those is a result rather than a correction. A latent model does not add a delta to a cell, it *reconstructs* the cell and adds a delta to the reconstruction, returning $P(C) + W\delta_z$. Referred to its own decoded baseline the reconstruction term cancels and the induced cosine is exactly 1.000000, as the theorem requires. Referred to the *real* control cells, which is what a retrieval scorer necessarily does, it carries each subpopulation’s reconstruction bias, which is cell-state dependent and does not cancel, and the model then appears to induce a great deal of divergence while inducing exactly none. **A distributional score run on such a candidate population is confidently scoring an autoencoder’s artefacts**, and that includes the structure diagnostics reported above whenever they are taken against real controls.

Gate 2, recoverability. Even where differential structure exists, a decision layer must be able to resolve it. We test this on a controlled mixture with ground-truth labels (K562 at 10 μ M, HDAC-class versus JAK-class cells), where the bimodal structure exists by construction. The question that matters is not whether clustering succeeds but *why it fails*, and those are two very different papers: if the information is absent from the representation no method can help, whereas if it is present and only the unsupervised assignment step cannot reach it, the bottleneck is an algorithm and is solvable. Distinguishing them requires an upper bound, which prior work in this area, ours included, has not reported. We therefore score every method in one unit, best-permutation accuracy against the true labels, with label matching given to each clusterer for free (Fig. 6e).

Unsupervised recovery is poor, the best of four methods reaching 0.674, but the informative number is the **supervised ceiling**, which reaches only **0.692** (AUC 0.752); a nonlinear learner does not beat a linear

one, so the boundary is not a hidden nonlinearity our probe was too weak to find. **The gap between the ceiling and the best unsupervised method is +0.018 accuracy**, so *within this benchmark* clustering already extracts essentially everything the representation holds and the limit is informational rather than algorithmic. Two later results retract almost everything one might conclude from that sentence, and we flag them here rather than let a reader carry it forward. The gap is an *underestimate*, because the design gives the clusterers the easier transductive representation and the probe the harder in-fold one (we had stated that bias backwards; Methods). And the 0.692 ceiling is largely an artefact of *pooling* several HDAC and several JAK compounds into two classes: split one drug against one drug, the same cells give 0.879 (Fig. 5d). The informational reading survives neither correction, and it does not survive contact with tissue. It is nonetheless a measurement rather than an insensitive probe: raising the separation between the two states artificially makes the ceiling pull away from clustering exactly as it should (Fig. 6f), and across 529 (cell line, drug) pairs in SciPlex3 the median silhouette of a two-way split is 0.034 (Extended Data Fig. 3).

But this benchmark is one we built, and the next section shows that it misled us. An HDAC-treated and a JAK-treated K562 cell are a mixture we assembled; they are not a subpopulation structure that occurs in a patient. Read on its own, the number above would license the conclusion that real subpopulations sit at an information ceiling no method can exceed, and we drew exactly that conclusion before testing it. It is wrong.

What Gate 2 does and does not constrain. Energy distance, MMD and sliced-Wasserstein are *cluster-free*: they compare whole populations and never form a partition, so Gate 2 is not a claim that they require one. The constraint is subtler, and it closes the loop with the functional-oracle result. A distributional distance sums every source of population difference into a single number, so where the subpopulation signal sits at a ceiling as low as the one measured here, that signal cannot be what dominates the distance. Something else must, and we have measured what: the candidate’s own response magnitude, which tracks the energy distance at $\rho = +0.791$. Gate 2 does not stop a cluster-free score from being computed; it determines whether the number that score returns is about subpopulation structure or about something else.

The two gates compound, and their consequences differ by score type. Note what Gate 1 is *not*: it is not structure preservation. Every predictor we tested preserves structure and every one still fails, so a generator validated on a structure-preservation metric would pass a test that does not bear on the question. Coverage-type scores, which need the partition explicitly, are blocked wherever Gate 2 is shut, which in this study is the constructed mixtures and not the patient tumours of the next section. Global scores are never blocked, but are left measuring whatever channel is largest, which is not the one their advocates intend.

Two further tests, both consistent with the criterion and neither rescuing it, are given in Supplementary Note 2: coverage performance rises with query identifiability but only weakly and not monotonically, and when queries are stratified by the Gate-2 variable itself the distributional advantage is flat.

HIR-Bench separates observable from circular predictors

To study these regimes under controlled conditions we construct HIR-Bench (Heterogeneous Inverse Retrieval Benchmark), a synthetic, method-neutral generator with a specified latent welfare oracle (Fig. 5). Each task instantiates one of three information conditions, observed, predicted_mean (candidates col-

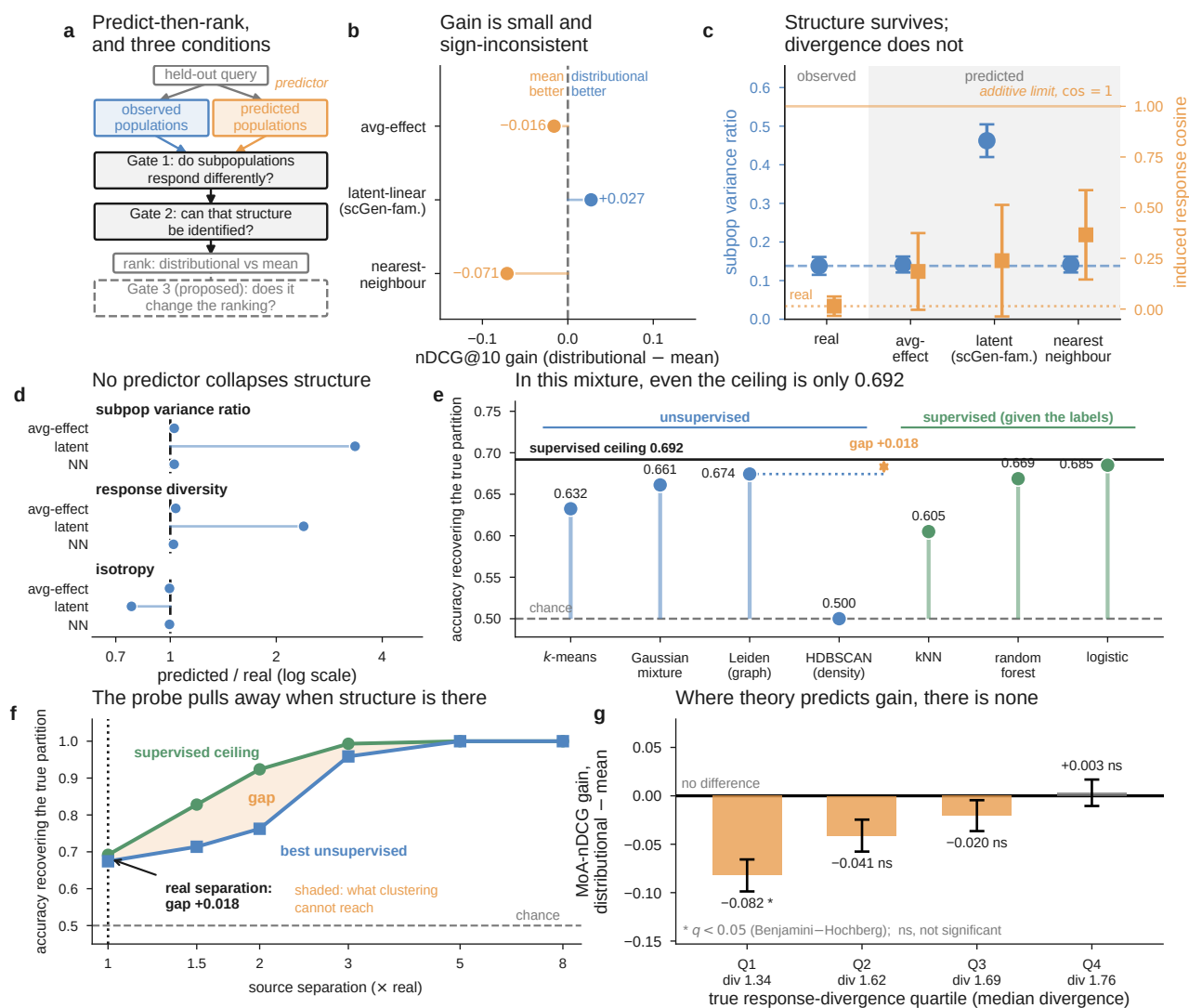


Figure 6: When can distributional information matter? Two measured gates, and a third condition they do not cover. Candidate populations must show differential response (Gate 1) and that structure must be identifiable (Gate 2); neither asks whether the difference changes the decision (Gate 3). **a**, The predict-then-rank design and the conditions a candidate population must satisfy: a held-out query is scored against candidate populations that are either observed or produced by a perturbation predictor. Gates 1 and 2 are measured in panels **b–f**. **Gate 3 is drawn dashed because it is proposed here rather than tested as the other two are**; its evidence is Fig. 5f and Fig. 4h,i, and it is what panel **g** is consistent with. **b**, Predictor-specific retrieval gain (nDCG@10, distributional minus mean). The gain is small and *not* consistent in sign: -0.071 (nearest-neighbour), -0.016 (average-effect), $+0.027$ (latent-linear); one task, so no dispersion is available. **c**, Gate 1. Predicted candidate populations *retain* baseline subpopulation structure (variance ratio 0.142 versus 0.138 for real ones, left axis) but do not reproduce *divergence*: the induced response cosine is 0.19–0.37 for predicted populations against 0.014 for real ones (right axis), where an additive predictor is pinned at exactly 1 by algebra. **The predicted values here do not measure that ceiling and we do not interpret them**, because these populations are cross-context blends and a latent model’s response against real controls also carries its reconstruction bias (Methods). **d**, Structure diagnostics, predicted relative to real. **e**, Gate 2 on a known bimodal mixture (K562, HDAC-class versus JAK-class cells), every method scored in one unit, best-permutation accuracy against the true labels. Unsupervised recovery is poor (best 0.674) and the *supervised ceiling* reaches only 0.692 (AUC 0.752), a gap of **0.018**, so *within this benchmark* clustering already extracts what the representation holds. That reading is superseded: the ceiling is largely an artefact of *pooling* drugs into classes, and in a patient’s tumour the same question gives 0.923 against 0.777 unsupervised, an algorithmic limit rather than an informational one (Fig. 5d). **f**, The control that licenses reading panel **e**: raising the separation between the two states makes the ceiling pull away from clustering as it should, so the absence of a gap at real separation is a measurement rather than an insensitive probe. **g**, The conditional advantage a working information condition would produce is absent. Distributional-minus-mean MoA-nDCG gain by *true* response-divergence quartile (bars are means, error bars s.e.m.; Benjamini–Hochberg corrected). No stratum is positive and the lowest-divergence quartile is significantly *negative* (mean -0.082 , $q = 5.8 \times 10^{-5}$, $n = 166$). Where the theory predicts the largest gain there is none.

lapsed to their means), and predicted_structure (candidates with partially preserved structure), and the generator admits an analytical flip boundary $\alpha^* = B/(A + B)$ separating the regime where distributional retrieval can help from where it cannot (Fig. 5a).

The grid comprises 13,440 benchmark *instances* (672 parameter configurations $\times$ 20 seeds; these are synthetic retrieval instances, not biological cells). The sanity checks hold (no-conflict flip rate 0.0000; median boundary margin 0.27; Extended Data Fig. 6d), and collapsing candidates to their means inflates decision regret most severely against the worst-welfare objective, the one that depends on the minority subpopulation (0.29 observed versus 0.96 predicted-mean; Extended Data Fig. 6c).

Is retrieval failure predictable, and from what? The second half of that question matters more than the first, and a benchmark with a latent oracle is the only place it can be answered cleanly. Features derived from the oracle are available to the benchmark’s authors but never to a retrieval method at query time; features of the observed populations are available to both. Fitting the same model on each feature set separates a usable diagnostic from a circular one, and the two answers are opposite.

We therefore run a 2×2 design: two feature sets crossed with two cross-validation groupings. The *observable* features (subpopulation-variance ratio, isotropy and response diversity of the query and candidate populations) are computable at query time. The *oracle-derived* features (top- k disagreement, Kendall conflict) are each a function of the benchmark’s latent utility matrix, as is the label. The groupings differ in their unit: the label is a deterministic step function of $(\alpha, \text{conflict})$, so the 13,440 instances contain only 28 independent parameter cells, whereas the natural leave-one-instance-out scheme treats 672 as independent and leaves roughly twelve near-identical copies of each held-out instance in training.

All four cells are shown in Fig. 5b. Read across the top row, retrieval failure *is* moderately predictable from what a method can see: AUC **0.788** against a majority-class rate of 0.643, the query’s response diversity carrying the highest weight. Read across the bottom row, the oracle-derived features are no better than chance (0.400; with 28 independent cells this is not distinguishable from 0.5). They predict one function of the latent utility matrix from another, which measures circularity, not predictability.

The two columns separate the two failure modes cleanly, and the interaction is the point. Pseudo-replication inflates the circular feature set by +0.240 (0.400 to 0.640) but the honest feature set by only +0.048 (0.788 to 0.836). An instance-level holdout preferentially rescues features that carry no generalizing signal, because memorizing near-duplicates is all they have; features with real signal barely benefit. A benchmark whose cross-validation unit is finer than its label-determining axes will therefore manufacture apparent predictability precisely where there is none, and the effect is largest for exactly the features an author is most tempted to use. Neither effect is specific to this benchmark.

Two caveats bound the 0.788. The effective sample size is 28 parameter cells, so a confidence interval computed on $n = 13,440$ would be meaningless, and by the same token the gap between 0.400 and 0.5 is not itself interpretable. And the label is a deterministic step function of those parameters, which makes every held-out fold single-class, so no per-fold AUC distribution exists and the pooled out-of-fold AUC is the only statistic available. “Moderately predictable” is as strong a claim as this supports.

Two limitations are essential. HIR-Bench is intentionally constructive: built from a specified latent welfare model, its phase boundary is expected by design, and the question that matters is whether real data project onto the same regimes. They do not: the boundary fails to transfer, scoring 61.1% against a 95.4% majority-class baseline (Methods). And one apparently supportive number is not evidence: the

gate “correctly” flags mean-collapsed predicted responses as no-distributional in 100% of the transfer test (the label the gate emits is no_DART, a pre-rename identifier retained in the released tables), but that rate is a tautology, since the rule returns it unconditionally for that information condition (Supplementary Note 1). HIR-Bench is a controlled framework for exposing failure modes, not a real-world diagnostic.

In real tissue Gate 2 is an algorithmic limit

Every real-data positive in this study so far lives in cell-line mixtures that *we* constructed. A minority subpopulation assembled by pipetting A549 together with K562 is not evidence about the heterogeneity a tumour actually has, and a criterion validated only on such a mixture is a criterion validated on our own pipetting. We therefore tested the two gates on ZhaoSims2021, acute-slice culture and surgical biopsy from ten glioblastoma patients, in which malignant glioma cells, tumour-associated myeloid cells and oligodendrocytes co-exist inside one patient’s tumour. Nobody mixed anything. Compartments are assigned by validated marker score and cells without a confident call are dropped rather than reassigned; an earlier attempt that took a bare argmax over five marker sets fabricated a 22% “T-cell” compartment whose cells did not express CD3 (Methods). This is *not* a retrieval benchmark and we do not use it as one: the panel is six drugs and only one patient received more than two.

Gate 2 must be measured on the same construct in both settings, and our first attempt was not (Fig. 5d). The constructed Gate-2 benchmark separates HDAC-treated from JAK-treated K562 cells: one lineage, cells differing only in which drug they saw, a *drug-response* partition. Our first natural test separated malignant from myeloid *control* cells: two lineages, a *cell-type* partition, and one that is trivially easy because a glioma cell and a macrophage differ in thousands of genes. It reached a ceiling of 0.964, and comparing that to 0.692 compared two different questions. Gate 2 is defined on the partition a retrieval score must resolve inside a candidate population, which is a drug-response partition. We therefore re-ran it as one: cells treated with drug A against cells treated with drug B, *within one compartment, within one patient*, which is the K562 construct transplanted into tissue nobody assembled (36 drug-pair splits; Methods). We additionally report the ceiling under the representation the clusterers actually receive, because the in-fold representation used earlier deflates the ceiling and so *shrinks* the very gap the conclusion rests on.

Two things follow, and the second one reverses us. First, the constructed ceiling was substantially an artefact of *pooling drugs within a class*: run at this comparison’s own granularity and representation the pooled construct gives 0.700, against the 0.692 the earlier in-fold run of the same construct reported. Splitting the same K562 cells one drug against one drug, as the natural test necessarily does, lifts the ceiling to 0.879 with unsupervised clustering at 0.837. Much of what we read as an information limit was the within-class heterogeneity of several distinct compounds we had merged.

Second, and this is the finding: in a real tumour the information is plainly *present*, and unsupervised clustering *cannot reach it*. The supervised ceiling is 0.923 while the best unsupervised method reaches only 0.777, a gap of +0.117 against +0.007 on the like-for-like constructed split. Centroid-based methods collapse (*k*-means 0.638, Gaussian mixtures 0.620) where the graph-based method partly survives (Leiden 0.780), which is the signature of a dominant variance direction that is not the drug-response axis: inside a real compartment, cell-state heterogeneity outweighs the drug response and unsupervised partitioning locks onto it.

We therefore withdraw the information-limit claim, and we withdraw the first withdrawal too.

625 The original conclusion, that real subpopulations sit at an information ceiling no method can exceed, is
626 wrong. But so is the reading we replaced it with, that real subpopulations are simply easy to resolve;
627 that came from the cell-type comparison and does not survive being posed as a drug-response question.
628 In real tissue Gate 2 is an *algorithmic* bottleneck, not an informational one: the partition is there to be
629 found, off-the-shelf clustering does not find it, and the gap is therefore something a better method could
630 close. That is the one place in this study where the news for distributional retrieval is constructive rather
631 than cautionary, and it is the opposite of what we concluded from our own benchmark twice.

632 **We attacked that gap three ways before believing it, because it is the only constructive claim in**
633 **this paper.** It is a difference of two *maxima*, so we recomputed every fixed (probe, clusterer) pair with no
634 selection anywhere; the 36 splits are not independent, since 30 come from one patient, so we bootstrap
635 over *patients* rather than splits, the pseudo-replication this paper criticises elsewhere; and we dropped
636 the dominant patient entirely. The gap survives all three, and dropping that patient makes it slightly
637 *larger*, not smaller (values in Fig. 5d; none of 5,000 bootstrap draws lies at or below zero).

638 **Three limits nonetheless bound it.** A cluster bootstrap with only *four* clusters under-covers, so the
639 interval is a floor on the uncertainty rather than an estimate of it. *Cells were dropped:* 39.9% fail the
640 compartment-assignment margin and are excluded rather than reassigned (Methods, Supplementary
641 Table 3), and since the discarded cells are by construction the hardest to place, both the ceiling and the
642 clustering accuracy are optimistic. *The drug panels differ:* the constructed split contrasts HDAC with JAK
643 inhibitors and the natural one six unrelated compounds, so response magnitude is not matched across
644 the arms. We report the comparison because it is the closest to like-for-like these data permit, and claim
645 no more precision for it than four patients and one drug panel can support.

646 **Gate 1 opens too, and the same benchmark misled us in the opposite direction** (Fig. 5e). Naturally
647 co-existing compartments do respond in different directions to the same drug, but far less so than our
648 mixtures implied: the induced response cosine has a median of 0.566 across 17 patient-drug pairs, where
649 our constructed mixtures sit at 0.014–0.044, essentially orthogonal. One drug pushes two unrelated cell
650 lines in unrelated directions; malignant cells and tumour-associated macrophages in one patient share
651 most of their response. The constructed benchmark overstates the divergence a distributional score
652 can exploit, by roughly twofold on a divergence scale ($1 - \cos$: 0.99 against 0.43), while misplacing
653 the identifiability limit. (The ratio of the cosines themselves, 40-fold, is not a meaningful measure of
654 divergence when one cosine is near zero.) Both distortions flatter this paper’s premise, and neither was
655 intentional.

656 **Both gates are therefore open in real tissue, and the two-gate account predicts that a distributional**
657 **advantage should follow. The mechanism that would produce one is missing, and its absence is**
658 **what Gate 3 names.** The paper’s motivating scenario, that two drugs can share a mean signature and do
659 opposite things to a subpopulation, can be tested directly on a real tumour. Across 18 within-patient drug
660 pairs, the similarity of two drugs’ *mean* signatures ranks the similarity of their *malignant-compartment*
661 responses at Spearman $\rho = +0.878$ (Fig. 5f). Where the mean says two drugs are alike, the malignant
662 compartment mostly agrees. Of the five drug pairs most similar on the mean, two have a malignant
663 response meaningfully less similar than their mean suggests, so the residual is real; it is also small.

664 **That statistic has a mechanical component, and we report the corrected one alongside it.** The mean
665 signature is taken over all called cells and malignant glioma is 41,314 of the 96,225 of them, so the two

quantities being correlated share 43% of their cells and part of the agreement is arithmetic rather than biology. Recomputed on compartments that share *no* cells, ranking the malignant-compartment response by the *myeloid*-compartment response instead of by the mean, the correlation is $\rho = +0.835$ (+0.878 to +0.835; on PW030 alone, +0.821 to +0.804). The finding survives the correction and the number moves, which is the honest way round. *We treat the disjoint value as the load-bearing one*, because it is the one that answers the question actually posed: whether what the bulk of the tumour does already tells you what the compartment of interest does. A paper arguing that a scoring rule and its evaluator must not share their object cannot exempt its own premise statistic from the same rule.

This is as close to a direct reading of Gate 3 as these data allow: where the mean already ranks what the subpopulation does, the differential response has little left to change about the decision. The two limits below are what stop it from being a measurement of Gate 3 rather than an indication of one.

Two limits on that number, and they are severe. It is close to a *single-patient result*: 15 of the 18 pairs come from PW030, the only patient given more than two compounds, and reuse the same six drugs, while three patients contribute one pair each. A p -value computed as though the 18 pairs were independent would be anticonservative by a wide margin, so we do not quote one; the honest description is a strong association in one patient, consistent in sign with three single-pair observations elsewhere. Testing the premise properly needs a cohort in which many patients receive many drugs, which this dataset is not. And a rank correlation between two similarity scores is neither a fraction of variance nor a statement that the response vectors are collinear, and we read it as neither.

We did not run retrieval here, so we report no retrieval result. ZhaoSims2021 supports six compounds with only one patient receiving more than two, and a leave-one-drug-out Hit@1 computed on that would be an anecdote wearing a benchmark's clothes. What this section shows is that the *mechanism* a distributional advantage would run through, mean-similar drugs diverging inside a subpopulation, is largely absent in real tissue: an inference about why an advantage should not be expected, not a measurement of its absence.

We then re-measured all three conditions at scale, on material that is neither ours nor tissue. Every gate number above rests on mixtures we assembled or on a cohort in which one patient supplies most of the pairs, and neither weakness can be fixed with these data. Tahoe-100M supplies a third kind of material: a complete grid of 50 cancer cell lines against 93 compounds at one dose, a vehicle arm in every line, a median of 648 cells per condition, and within-line heterogeneity that is intrinsic cell state rather than two populations pipetted together (plate 3; Methods). It is not tissue and we do not use it to argue relevance. We use it for sample size, and we recompute the same three statistics rather than new ones (Extended Data Fig. 7).

Gate 1 widens the distance between constructed and unconstructed material. Splitting each treated population by cell-cycle phase, each subpopulation referred to its own phase-matched control, the induced response cosine over 3630 conditions has a median of 0.739 (interquartile 0.600 to 0.836), and only 0.11% of conditions are as divergent as our constructed mixtures. The mixtures are not at one end of the natural range; they are outside it. The exclusion rule is mildly adverse here and we checked it: conditions are dropped when a subpopulation arm falls below 50 cells, which preferentially removes cell-cycle-arresting compounds, and those do respond more divergently, but the effect is negligible (Spearman +0.077 between a drug's coverage and its cosine; the median moves from 0.739 to 0.740 on well-covered

707 compounds alone).

708 *Gate 2 replicates, and the constructed benchmark is again the outlier.* Posed as the same drug-response
709 question and scored with the same four-method clusterer panel, the supervised ceiling is 0.853 and the
710 best unsupervised method reaches 0.611: a gap of 0.157 over 960 drug pairs, positive in 48 of 48 cell lines.
711 No method family rescues it, and the pattern differs from tissue in an informative way: k-means 0.592,
712 Gaussian mixtures 0.582 and Leiden 0.563 all sit near the 0.5 null, where in the tumour it was Leiden
713 that partly survived. The information is present and off-the-shelf partitioning does not reach it, which is
714 what the single tumour cohort suggested and could not establish.

715 *Gate 3 replicates and is qualified, which is the more useful outcome.* Ranking one compartment's response
716 similarity by another's, on disjoint cells, gives a median Spearman of 0.841 over 44 cell lines under the
717 cell-cycle split, and 0.778 over 45 under a partition fitted on control cells and fixed before any treated
718 cell is seen, against +0.835 for the tumour computed the same way. It is *not* universal: under the control-
719 state partition 2 of 45 contexts fall below 0.5, the lowest at 0.236. Those are contexts in which the bulk
720 does not tell you what the minority does, and they are where a distributional score should pay off. A
721 condition that is usually met and sometimes not is a condition; one that is always met would have been
722 a verdict. Nor is Gate 3 Gate 1 restated: within the cell-cycle partition the two correlate across contexts
723 at Spearman 0.620 ($R^2 = 0.385$, $n = 44$), and where subpopulations respond most differently (Gate 1
724 cosine 0.394 to 0.534) Gate 3 does fall, to 0.605 to 0.764, without reaching 0.5.

725 **Four limits bound what this adds, and the first is the one that matters.** These are cell lines. Their
726 heterogeneity is unconstructed, which is the property the argument needs, but it is not a tumour, and
727 the paper's claims about relevance continue to rest on tissue. Second, this is one plate of fourteen, 93
728 of the 380 compounds in the resource. Third, both partitions are coarse two-way splits of cell state,
729 not the persist or resistant-clone axis that motivates this work, so what is measured is whether *a* real
730 subpopulation structure behaves as the conditions predict, not whether *that* one does. Fourth, which
731 contexts show an open Gate 3 depends on the partition: the two partitions agree at the extremes, and
732 the same two cell lines are the most open under both, but their rankings agree only moderately overall
733 (Spearman 0.466). Gate 3 can flag a handful of contexts; it cannot rank them finely until the partition is
734 fixed in advance.

735 **The two gates are necessary but not sufficient, and this is how we found out.** Opening them does
736 not deliver a distributional advantage, because in real tissue the compartment-specific response is mostly
737 ranked by the mean already. The independent functional oracle reaches a compatible conclusion by an
738 entirely different route: energy retrieval beats the mean incumbent there, and a scalar magnitude control
739 then reproduces almost the whole of that margin. The two lines of evidence share no data, no metric
740 and no code path, and they do not share a number either, so we do not claim they converge on one. Both
741 are consistent with Gate 3 being the binding one here, and neither was designed to test it. What they
742 agree on is a direction: distributional information in this domain is real, positive, and much smaller than
743 objective-aligned evaluation makes it appear.

744 Discussion

745 The main result of this work is not that the distributional score is better or worse, but that *how much*
746 better it is depends almost entirely on who is asked. Under a metric of its own distributional kind

747 the advantage is large. Under an independent, semantically matched functional oracle it is real but
748 small, and almost all of what remains is a response-magnitude channel that a scalar reproduces without
749 comparing any distributions at all. Under a task-proximal mechanism criterion it is absent. The same
750 rankings, the same scorer, the same queries: only the evaluating criterion changes. The score family
751 released here is valuable as a probe, an instrument that can be scored by its own objective and by more
752 independent criteria on the same tasks, rather than as a therapeutic recommender. The paper's positive
753 anchors survive intact: the unification of mean and distributional retrieval is an algebraic fact, and there
754 is genuine distributional signal in response-available populations. What does not survive is the size of
755 the gain that objective-aligned evaluation reports for it.

756 We also asked, as a direction for future work rather than a load-bearing result, whether the re-
757 sponse divergence a distributional score measures has any biological correlate. In the Frangieh Perturb-
758 CITE-seq data, cells whose response direction diverges most between contexts are weakly enriched for
759 resistance-associated programs. Without a drug timecourse, a survival readout or a post-treatment clone
760 measurement this is hypothesis-generating only, we make no claim that the distributional score predicts
761 resistance, and the analysis is reported in Supplementary Note 3 and Extended Data Fig. 4 rather than
762 here.

763 **Why the negative results are central**

764 The predict-then-rank nulls and the independent-metric results are the evidence that supports this pa-
765 per's core claim, not a shortfall against it. The objective-aligned positive on its own would license the
766 conclusion that distributional retrieval substantially improves ranking. The task-proximal metrics do
767 not support that, and the independent functional oracle supports a far weaker version of it: the distri-
768 butional score does beat its incumbent there, but a scalar that performs no retrieval reproduces nearly
769 the whole margin. The positive and the null together, labeled by metric class, are what tell a practitioner
770 which conclusions a given metric licenses and which it does not.

771 **Implications for virtual-cell and perturbation-prediction models**

772 The two-gate result has a direct implication for the fast-growing class of perturbation predictors and
773 virtual-cell foundation models. In this framework such models are candidate-response sources, and
774 whether a distributional decision layer can add anything on top of them depends entirely on whether
775 they supply *differential* response. The load-bearing statement here is algebraic and needs no benchmark
776 to establish: a predictor that adds one delta vector to every cell induces exactly zero response divergence,
777 so additivity is *sufficient* to make a distributional decision layer worthless on top of it. That is a property
778 of a model class, not a deficiency of any implementation, and no amount of training removes it.

779 What we do not claim is that today's predictors are, as a field, additive. We can say it of one deployed
780 model: the published scGen variational autoencoder passes its learning check and sits at a true induced
781 divergence of 0.972 against the additive ceiling of 1.000, reproducing about 3% of the divergence real
782 subpopulations show. That is a fact about scGen, not about the field. The in-repository linear-latent
783 model is additive by construction, so its value demonstrates the algebra rather than surveying anything.

784 Non-additive, cell-state-conditional predictors exist, including optimal-transport, flow and
785 Schrödinger-bridge formulations^{17,18}, and a latent-space optimal-transport map does open Gate 1,
786 though only a crack. What we have not done is the second half of the test. A predictor that genuinely

787 produces state-specific responses should, if the two-gate framework is right, restore a distributional
788 advantage that additive predictors cannot, and one that opens Gate 1 yet yields no such advantage
789 would falsify the framework. That test is out of reach for the predictors we could run: the optimal-
790 transport map’s cross-line predictions are too weak to retrieve with, and the flow model this test most
791 plausibly calls for, CellFlow²⁵, learns and opens Gate 1 within a context but does not carry that learning
792 to a held-out cell line, because per drug the transfer is bounded by the drug’s cross-context response
793 similarity and the retrieval task selects exactly the drugs for which that similarity is near zero (Results).
794 The decisive test therefore still requires a predictor that both predicts accurately *across contexts* and
795 opens Gate 1, most plausibly a flow or Schrödinger-bridge model trained to transfer; we name this as
796 the test that would falsify us, and record that the flow model we trained does not yet complete it.

797 The methodological point stands independently of that test, and it is sharper than we first realised.
798 Differential response, not mean-level accuracy and not even structure preservation, is the property that
799 determines whether distributional retrieval is applicable at all. Structure-preservation metrics alone
800 would have scored every predictor we tested as adequate. Worse, they would have scored the gener-
801 ator’s own reconstruction error as biological structure. An additive latent model induces exactly zero
802 true response divergence, yet measured against real control cells, as a scorer must, it appears to in-
803 duce a great deal, because the autoencoder reconstructs the two subpopulations with different biases.
804 A distributional decision layer stacked on such a model would be scoring the autoencoder. Anyone
805 reporting that a generative perturbation model “preserves subpopulation structure” or “induces diver-
806 gence” should therefore state which baseline the claim is measured against, and show that it survives
807 being measured against the model’s own output. This is consistent with independent benchmarking,
808 in which foundation models and deep-learning predictors did not outperform deliberately simple base-
809 lines for perturbation prediction^{11,27}: objective-aligned reconstruction accuracy need not translate into
810 an advantage over mean or linear baselines.

811 **When distributional information can matter**

812 The question a practitioner actually asks is not whether distribution-aware retrieval is defensible in prin-
813 ciple but when it is worth reaching for, and the results here answer it as three conditions that must hold
814 together:

815 *heterogeneous responses exist* (Gate 1, differential response) + *those responses are recoverable* (Gate
816 *2, recoverability*) + *they change which candidate is preferred* (Gate 3, decision relevance)

817 Each is necessary and none is sufficient, which is the practical content of the result: we found a setting,
818 patient glioblastoma, in which the first two hold and no advantage follows. The three are not on equal
819 evidential footing and we do not present them as though they were. Gates 1 and 2 are measured here, on
820 constructed mixtures and on patient tumours, and Gate 1 in addition has an algebraic form that needs
821 no benchmark: an additive predictor shuts it by construction. Gate 3 is proposed rather than measured,
822 and rests on evidence gathered to answer other questions, chief among it a within-patient correlation
823 dominated by a single patient (Results). A study reporting a distributional gain can nonetheless check
824 all three, and the third is the cheapest of them: show that the ranking the distribution produces differs
825 from the ranking the mean produces on the outcome being claimed. Where it does not, the distribution
826 is being computed but is not deciding anything.

827 What independent validation would require

828 The external functional readout we could reach, GDSC dose-response profiles matched to our drug
829 panel, constrains the question without settling it. It is a semi-real oracle: viability is measured in a
830 separate bulk assay, matched by compound name, under a different dose regime. It establishes that tran-
831 scriptional similarity recovers functional similarity better than a mean signature does, and that almost all
832 of that recovery is a magnitude channel. It cannot establish therapeutic utility. Turning this framework
833 from an evaluation probe into a therapeutic method would require readouts measured on the same cells
834 the method is asked to treat: dose-response viability *in situ*, longitudinal treatment survival, functional
835 perturbation validation, post-treatment clone expansion, and combination-response validation. Until
836 then, the defensible claim is that distributional information is *present* and *modestly useful*, and far smaller
837 than objective-aligned evaluation makes it appear.

838 We urge four specific controls on anyone reporting a validation of a distributional retrieval method,
839 each of which caught an error of ours.

840 First, match the oracle to the task. A similarity retriever graded on absolute potency is being asked
841 a question it was never posed, and it will fail in a way that looks like a finding: potency tracks response
842 magnitude, a distributional distance also tracks response magnitude, and the two confound. We ran that
843 comparison, drew a conclusion from it, and had to withdraw the conclusion; it now appears only as an
844 audit.

845 Second, beat the magnitude scalar. Rank the candidates by how closely their response magnitude
846 matches the query's, compare no distributions at all, and show that your method beats it. On the
847 matched oracle ours does not do so significantly, and on the mismatched one it loses outright. A dis-
848 tributional method that cannot clear this bar has not shown that the distribution is what is doing the
849 work.

850 Third, vary the shape of your external oracle, not just its identity. It is natural to assume that once
851 a criterion is external, independent and invisible to the method, it can be trusted to arbitrate. It cannot.
852 Two oracles built from the same proteins in the same cells, one computed as a mean and one as a distri-
853 bution, hand victory to opposite methods (Fig. 5c). A field that validates distributional methods against
854 distributional readouts and mean-based methods against mean readouts will find both vindicated, and
855 neither finding will be about biology. The practical form of this control is cheap: compute your external
856 criterion in at least two statistical forms and report both. If your method only wins under the form that
857 matches its own, say so.

858 Fourth, and least comfortable, do not let a benchmark you built decide a claim about biology, and
859 check that the two things you are comparing are the same question. Our constructed cell-line mixtures si-
860 multaneously understated how identifiable real subpopulations are (ceiling 0.879 on a like-for-like split,
861 against 0.923 in a patient's tumour, where our class-pooled mixture had said 0.692) and overstated how
862 divergently they respond (cosine 0.014 against 0.566). Both distortions happened to favour the premise
863 we were testing, and neither was visible from inside the benchmark. Worse, our first attempt to check
864 the benchmark against tissue compared a drug-response partition in the mixture with a *cell-type* par-
865 tition in the tumour, which is a far easier problem, and it produced a reassuring number (0.964) that
866 meant nothing. A comparison across two settings is only as good as the sameness of the question asked
867 in each, and it is alarmingly easy to hold the protocol fixed while silently changing the question. Any

information-condition criterion validated solely on synthetic or constructed heterogeneity should be treated as a statement about that construction until it is checked against tissue, and the check must ask tissue the same question.

Field-level evaluation implications

The broader lesson is that the central distinction is not distributional versus mean retrieval, but objective fidelity versus independent utility. Our field-level audit shows that objective-aligned evaluation is the default across signature retrieval, expression prediction, and graph-based inverse design: the headline metric is usually a restatement of the method's own objective. This is legitimate for testing objective fidelity, and we do not claim any prior method is wrong. But it means that strong benchmark numbers, on their own, answer a narrower question than they are often read to answer.

Two corollaries follow. A benchmark whose cross-validation grouping is finer than its label-determining axes will manufacture apparent predictability from nothing, as ours does for oracle-derived features^{28,29}; and a distributional score's apparent link to therapeutic value can be a response-magnitude confound, so any such claim should be required to beat a query-independent magnitude baseline. Neither failure mode is specific to this work. This reading is corroborated by recent metric-focused studies: the distributional metrics used here have documented failure modes, with the Wasserstein distance degrading under variance scaling in high-dimensional expression space and the energy distance overlooking disruptions in gene-gene dependencies²¹, and metric choice alone can flip model rankings²². We cite the first against ourselves; it bounds the interpretation of our own Class-A scores. The contribution is to make the objective-fidelity/independent-utility distinction explicit and measurable on a single task set, a template other inverse-retrieval methods could adopt.

The same distinction may matter outside retrieval, and we raise it as a question rather than as a result. Objective-utility mismatch as we define it is a property of retrieval evaluation, and it is the only place we have measured it. The ingredient it needs is not peculiar to retrieval, however: a surrogate objective standing in for a functional outcome nobody measured. Virtual-cell and perturbation-prediction models are scored on held-out reconstruction of the quantity they were trained to reconstruct, and whether that tracks downstream decision quality is a separate question that independent benchmarks have already begun to raise^{11,26,27}. Biomarker discovery selects features by an association statistic and is frequently validated by the same statistic in a new cohort, which tests transportability rather than clinical utility. Molecular design optimises a computed property and is commonly evaluated by that property recomputed. We have tested none of these and claim nothing about them. What transfers is the check, not the finding: report a result under at least one criterion the method's own objective did not define, and say which class the headline number belongs to.

Limitations and scope of claims

A full statement of what this study does *not* claim, seventeen items covering the scope of every result reported here, is given as Supplementary Note 4. The load-bearing ones are these. No gain in this study is measured on a clinical or same-cell endpoint. The Class-C positive is conditional on the cell-line centering of its oracle and does not survive the uncorrected construction, and under neither construction does the distributional score significantly beat a scalar that compares no distributions. We do not claim that the predictors tested here represent the field, nor that opening Gate 1 buys retrieval gain, since we

could not build a predictor that both transfers across contexts and opens it. And we do not report a count of distributionally dominant real tasks, because the conventional threshold sits inside the noise band of the difference distribution.

Three limits bound the scope of the whole study rather than any single result, and we state them as such.

First, the external oracle is imperfect, and its imperfection is not incidental to the argument. The Class-C readout is semi-real: viability measured in a separate bulk assay, joined to the transcriptional data by compound identity rather than by shared cells, under a different dose regime. A paper whose thesis is that the choice of criterion decides the winner cannot exempt its own criterion, and we do not: the affirmative Class-C result holds under the cell-line-centered oracle we argue is correct and reverses under the uncorrected one, and both are given in Supplementary Table 4. The oracle-shape test makes the same point sharper, since two external criteria built from the same proteins in the same cells hand victory to opposite methods.

Second, this study audits retrieval, not prediction. Every result here concerns how candidates are *ranked* and how that ranking is *graded*. Perturbation predictors enter only as candidate-response sources, and the one statement we make about a model class, that additivity shuts Gate 1, is algebraic rather than a benchmark of any implementation. Nothing here measures whether distributional *prediction* objectives improve prediction, which is a different question with its own literature^{17,18,25}.

Third, prospective validation remains outstanding, and it is what would settle the question. No endpoint in this study is measured on the cells a method is asked to treat. Converting the conditional claim made here, that distributional information is present and modestly useful, into a therapeutic one requires same-cell readouts: dose-response viability *in situ*, longitudinal treatment survival, and post-treatment clone expansion. Until such a readout exists, the affirmative content of this paper is bounded by the semi-real oracle that produced it.

Methods

Data sources and preprocessing

Three single-cell datasets carry every result reported here. SciPlex3³⁰: 276,325 cells $\times$ 2,000 genes across three contexts (A549, K562, MCF7) and 188 drugs, all present in all three contexts. Frangieh melanoma Perturb-CITE-seq³¹: 218,023 cells $\times$ 2,000 highly variable genes across Control, IFN γ , and Co-culture conditions with 239 CRISPR knockouts. CD34+ (GSE306429)³²: 33,984 cells $\times$ 2,000 genes, 36 drugs, used as an honest negative control. A fourth resource, the LINCS closed-loop / PDGrapher benchmark (41,070 rows across A549, MCF7 and PC3, 1,369 drugs), was prepared for a graph-based inverse-design comparison that we do not report; *no number in this paper is derived from it*, and we name it only because it ships with the code. A fifth, GDSC dose-response, is the external potency oracle used for the Class-C analysis and is described below. Expression is processed to log-normalized differential profiles against matched controls; per-context control means and treated-population rows are exposed through a common dataset interface. Dataset scale, preprocessing settings and query construction are tabulated in Supplementary Table 1.

946 EvalShift retrieval task and distributional metrics

947 Given a target population Q and candidate populations $\{P_d\}$, the retrieval score ranks by a population-
948 to-population distance. Implemented scores: the energy distance³³ (subsamped to a fixed maximum
949 cell count with a fixed seed), an RBF maximum-mean-discrepancy³⁴, a sliced-Wasserstein distance (128
950 projections)³⁵, and a subpopulation-coverage aggregate with a temperature parameter β interpolating
951 from mean aggregation ($\beta \rightarrow 0$) to worst-case ($\beta \rightarrow \infty$). Mean-signature retrieval, by cosine or by
952 Euclidean distance between mean signatures, is the zero-variance limit of the same family.

953 Signature and PCA baselines

954 CMap-style signature retrieval is implemented with two selectable scoring rules. The cosine rule com-
955 puts the cosine of mean-delta signatures and is the same operation as mean-cosine retrieval, which is
956 the source of the exact Hit@1 identity. The WTCS rule is a rank-based (GSEA-style) weighted connectiv-
957 ity score, a monotone sibling that is not algebraically identical. PCA-latent retrieval fits PCA per query
958 and scores in latent space by mean or distributional distance.

959 Predict-then-rank baselines

960 Candidate responses are generated by an average-effect predictor (per-drug mean deltas), a nearest-
961 neighbour predictor (nearest training context by control-state similarity), and a CPA-style linear-latent
962 predictor, used as a surrogate for the latent-arithmetic family that scGen⁵ belongs to; we do not bench-
963 mark a trained scGen, and the active backend is recorded in every output. They are used purely as
964 candidate-response *sources*, not as models under benchmark. They do not stand in for the broader class
965 of published perturbation predictors and virtual-cell foundation models (scGen⁵, CPA⁶, chemCPA¹⁴,
966 GEARS⁷, scGPT², scFoundation³, Geneformer⁴), and no claim about that class is drawn from them. An
967 earlier draft asserted that they did stand in for it; that is inconsistent with the algebraic scope of the
968 Gate-1 result and with limitation 1, and it is withdrawn.

969 **How a predicted effect becomes a predicted population.** A distributional score ranks *populations*,
970 so a predict-then-rank baseline must turn a predicted effect into a population, and that synthesis step
971 silently determines what structure the population can possibly contain. We therefore state it explicitly.
972 A predicted population is built by applying the predicted effect to the query context’s *real control cells*,
973 one cell at a time. This is what the latent-arithmetic models in this class do (encode observed cells, shift
974 each cell’s code, decode each cell), and it is the only construction under which a predicted population can
975 carry the control population’s covariance and any multimodality it has. The alternative and superficially
976 natural construction, drawing each predicted population as control mean + δ + iid $\mathcal{N}(0, \sigma)$, must not be
977 used for structure diagnostics: that cloud is unimodal by construction (isotropy index 0.998), so a k -
978 means $k=2$ between-total variance ratio computed on it returns ≈ 0.009 whatever the predictor learned.
979 Any structure diagnostic evaluated on such a synthesizer measures the synthesizer, not the predictor.
980 Both constructions are implemented behind an explicit flag so the difference can be reproduced.

981 **In-sample ceiling.** In the predict-then-rank experiment the predictors are fit on the full SciPlex3
982 matrix and the query is composed of cells from two cell lines that remain in the training set, so the
983 nearest-neighbour predictor can in principle transfer a signature it estimated from the very cells that
984 compose the query. The reported predict-then-rank numbers are therefore an in-sample ceiling, not a
985 transfer result. A leave-both-contexts-out option is provided, and we do not claim a leave-one-context-

986 out guarantee for the reported run. The direction of this bias favours the mean incumbent, so it does not
 987 manufacture the negative result reported here; it bounds it.

988 **Gate 1 measured within a context, and the calibration check that caught us**

989 **The theorem is a within-context statement.** An additive predictor adds one delta vector to every cell
 990 *of the query context*. Its two subpopulations therefore receive identical deltas and $\cos(d_0, d_1)$ is exactly 1,
 991 but only if the two subpopulations *are in that context*. Our first Gate-1 experiment measured the cosine
 992 on an alpha-blended candidate population whose two subpopulations are two different cell lines, and
 993 an additive predictor emits a different delta per cell line, so what it measured was $\cos(\delta_{A549}, \delta_{K562})$. That
 994 is the similarity of two contexts' deltas and is not a test of additivity.

995 **The calibration check is what exposed it, and it is asserted rather than inspected.** The experiment
 996 scores predictors that are additive *by construction*, whose value is fixed by the algebra at 1.000. They
 997 returned 0.267 and 0.353. A control that is required to hit a known value and misses it condemns the
 998 instrument, not the models, and the script refuses to emit numbers when it fails. We recommend the
 999 practice generally: an experiment whose theory pins a control to an exact value should assert that value,
 1000 not merely report it.

1001 **The corrected design** (analysis/predictors/gate1_within_context.py). For each context we take up
 1002 to 400 control cells and split them into two subpopulations by their own cell state (30-component PCA,
 1003 k -means with $k = 2$; subpopulation sizes and silhouettes are reported, because a low induced cosine
 1004 would mean nothing if the within-context structure were not there). For each drug the predictor is run
 1005 *separately on each subpopulation's control cells*, so the partition is true by construction and no assignment
 1006 step is involved:

$$d_k = \text{mean}(\text{predicted cells of subpop } k) - \text{mean}(\text{control cells of subpop } k), \quad \cos(d_0, d_1).$$

1007 Population synthesis draws a permutation of the supplied cells, which preserves a mean exactly. The
 1008 real treated cells of the same context, partitioned by the same control-derived subpopulations, give the
 1009 reference value. Gate 2's assignment error is no longer entangled with this measurement; it is measured
 1010 separately (above).

1011 **Two baselines, because a generative model does not return $C + \delta$.** A latent predictor returns $P(C) +$
 1012 $W\delta_z$, where P is its own reconstruction of the input. The response of a subpopulation therefore depends
 1013 on what it is referred to, and the two choices are not the same quantity:

$$\underbrace{d_k = \text{mean}(P(C_k)) - \text{mean}(C_k) + W\delta_z}_{\text{against the REAL control cells: what a scorer measures}}, \quad \underbrace{d_k = W\delta_z}_{\text{against the model's own decoded control}}.$$

1014 The first carries $\text{mean}(P(C_k)) - \text{mean}(C_k)$, the reconstruction bias of subpopulation k , which is cell-
 1015 state dependent and does not cancel; the two subpopulations' biases are themselves near-orthogonal
 1016 ($\cos = -0.05$ to -0.18). The second cancels it exactly. We report both. The additivity theorem is a
 1017 statement about the second, and the calibration check is asserted on it: an additive predictor must return
 1018 1.000000, and a run in which it does not is discarded. A predictor that does not reconstruct its input
 1019 (the gene-space average-effect predictor, and the optimal-transport map, which returns C plus a per-cell

displacement) has no baseline of its own, and for those two the columns coincide by construction, which is the correct answer rather than a bug. Which predictors reconstruct and which do not is declared explicitly in the code rather than inferred.

Information-condition diagnostics and gate diagnosis

The information condition is tested as two gates. Gate 1 (differential response) uses the predict-then-rank experiments: candidate responses are generated by each predictor family, ranked by the distributional score, and compared to mean retrieval, with structure diagnostics computed on generated versus real populations. Two quantities are reported. The *subpopulation-variance ratio* is the k -means $k=2$ between-total variance ratio of the candidate population. The *induced response cosine* is $\cos(d_{\text{maj}}, d_{\text{min}})$, where d_k is the mean of subpopulation k minus the mean of its own matched control population; it is the quantity a distributional score must exploit, and it is identically zero for any predictor that adds a single delta vector to every cell. The reference population for both is the alpha-blended candidate population that a scorer actually ranks, not a single-context treated population.

Gate 2 (recoverability) asks whether a query’s subpopulation structure can be resolved at all, on a controlled mixture with known two-state composition (K562 at 10 μM , HDAC-class versus JAK-class cells; 50–400 cells per source, 10 seeds). The design point is that clustering failure on its own is uninformative: it is equally consistent with the information being absent from the representation and with its being present but unreachable without labels, and those have opposite implications for whether single-cell resolution can ever pay off. The analysis therefore reports a supervised ceiling alongside the unsupervised methods, and the decisive quantity is the *gap* between them, not either number against chance.

Common unit. Adjusted Rand index and classifier AUC are not comparable, so every method is converted to best-permutation accuracy against the true labels: each cluster is assigned its majority true class. For $k = 2$ this is the better of the two permutations; for $k > 2$ it is strictly generous, since an over-partitioning method is never penalised for splitting a true class. HDBSCAN noise is treated as its own cluster and likewise assigned its majority class, which is generous again. Clusterers thus receive their label matching for free.

Unsupervised methods: k -means and Gaussian mixtures, both handed the true $k = 2$; Leiden³⁶ (graph-based, resolution 1.0, which selects its own k); HDBSCAN³⁷ (density-based, which selects its own k and may return none). The last two are the field’s actual defaults and were absent from our earlier analysis; adding them does not change the answer. All four see the same PCA representation (30 components) fit on the full mixture, which is the transductive setting an unsupervised pipeline would in fact have.

Supervised ceiling: logistic regression, random forest and k -nearest-neighbours, each given the ground-truth labels under 5-fold stratified cross-validation, with the PCA *re-fit inside each training fold* and test cells only projected through it, so no test-fold covariance informs the representation. The ceiling per run is the best of the three; including a nonlinear learner is what distinguishes an information limit from a too-weak-probe artifact.

The direction of the resulting bias, corrected. An earlier version of this Methods section stated that giving the clusterers the easier transductive representation while making the probe re-fit its PCA in-fold biased the comparison *against* the conclusion drawn. That is backwards. The conclusion drawn is that the *gap* is small; inflating the unsupervised arm and deflating the supervised arm makes the *gap smaller*, so the design flatters that conclusion rather than challenging it. We therefore also report

1061 the ceiling under the *matched* representation, the same transductive PCA the clusterers receive, which
1062 leaks no labels because PCA does not see them, and it is the matched gap that should be read (analysis/
1063 sis/natural/gate2_drug_response.py).

1064 *Positive control.* Source separation is scaled by a factor $s \in \{1, 1.5, 2, 3, 5, 8\}$ using a transform fit once
1065 on the full source populations, never on the sampled cells; $s = 1$ is the real data. Verifying that the
1066 ceiling pulls away from clustering as s rises is what licenses reading the absence of a gap at $s = 1$ as a
1067 measurement rather than as an insensitive probe.

1068 Query identifiability on real drugs is the silhouette of a two-way k -means split computed per drug
1069 on the query cells alone.

1070 The gate is additionally audited directly by re-running the partial-observed retrieval with an addi-
1071 tively computed true response divergence column (leave-out-free, from the observed candidate popula-
1072 tions), then correlating each gate axis with true divergence and stratifying retrieval gaps by divergence
1073 quartile with Benjamini–Hochberg control and power analysis. This audit is the basis for the findings
1074 that the structure-reliability axis is anti-correlated with true divergence and that the minority-coverage
1075 effect, while significant, is negligible.

1076 **Dataset positions on the divergence gate.** For each dataset the response cosine between the two
1077 subpopulations of a query is computed from the observed populations with the same probe used inside
1078 the gate, giving 0.032 (cross-line SciPlex3), 0.709 (Frangieh) and 0.186 (CD34+).

1079 **HIR-Bench generator and synthetic welfare oracle**

1080 HIR-Bench generates heterogeneous retrieval tasks from a specified two-subpopulation latent welfare
1081 model with tunable conflict, yielding an analytical flip boundary $\alpha^* = B/(A + B)$. Each task is instan-
1082 tiated under the observed, predicted_mean, and predicted_structure information conditions. Reported
1083 numbers come from the full 13,440-*instance* grid (672 parameter configurations $\times$ 20 seeds; $n_+ = 8,640$,
1084 $n_- = 4,800$).

1085 **The predictability model.** A logistic regression predicts retrieval flips from two feature sets, fitted
1086 and reported side by side because the contrast between them is the result. The *observable* set contains only
1087 quantities a retrieval method can compute at query time: the subpopulation-variance ratio, isotropy
1088 index and response diversity of the query population and of the candidate populations. The *oracle-*
1089 *derived* set contains top- k disagreement and Kendall conflict scores, each of which is a function of the
1090 benchmark’s latent utility matrix, as is the label.

1091 Folds are held out one *label-determining parameter cell* at a time, that is, one $(\alpha, \text{conflict})$ pair, because
1092 the label is a deterministic function of those two axes; there are 28 such cells and this is the effective
1093 sample size. Grouping by the instance identifier instead, the natural choice, is invalid here: that identifier
1094 also encodes axes that leave the utility matrix unchanged, so each held-out instance leaves roughly
1095 twelve near-identical copies of itself in the training set. We report the AUC under that grouping (0.640 for
1096 the oracle-derived features) only to quantify the inflation it produces. Because the label is a deterministic
1097 step function of the grouping variables, every held-out fold is single-class, no per-fold AUC is defined,
1098 and the pooled out-of-fold AUC is the only available statistic.

1099 Partial-observed retrieval protocol and real-data projection

1100 The partial-observed protocol scores candidates by an energy-based welfare regret against the latent
1101 objective and records both the objective-aligned regret (Class A) and task-proximal metrics (Class B:
1102 MoA-recovery nDCG³⁸, minority-state coverage) per query.

1103 **Held-out unit, and what it does not license.** The held-out unit in the gate audit is a cell line (A549,
1104 K562, MCF7), and all three come from SciPlex3, so they share batch, protocol, highly-variable-gene selec-
1105 tion and normalization. This is leave-one-cell-line-out within a single experiment, not leave-one-dataset-
1106 out, and we make no cross-dataset generalization claim on its basis.

1107 **Undefined metric values are excluded, not encoded.** MoA recovery is *undefined* for the leave-MoA-
1108 out and partial-library splits, where the entire mechanism class is removed from the candidate library,
1109 so no correct answer is present to recover. Those 165 of 765 queries (21.6%) are dropped, giving $n = 600$.
1110 They must not be carried as a sentinel value: differencing a sentinel against itself yields a structural zero
1111 that enters every stratum as a spurious no-effect observation and pins the median gap at exactly 0.000.
1112 Excluding them does not rescue the metric; it makes the negative result sharper.

1113 **Real-data projection.** The projection maps the four datasets onto the HIR-Bench regime plane and
1114 records, per task, the minority-state coverage of the distributional pick minus that of the mean pick. The
1115 full configuration builds 239 tasks, and the composition is given here because the three-factor description
1116 in an earlier draft accounted for only 180 of them: 90 constructed cross-line tasks (three cross-line pairs
1117 $\times$ ten drugs $\times$ three seeds), 90 within-line (three cell lines $\times$ ten drugs $\times$ three seeds), 23 Frangieh, 12
1118 CD34+ and 24 under predicted candidates. We report the distribution of that difference.

1119 **Why the dominance count is withheld.** A task is conventionally called distributionally dominant
1120 when the difference exceeds $+0.01$. We compute the count and do not report it as a finding, for three
1121 reasons. (i) The threshold is not calibrated to the data: it sits at 0.77 standard deviations of the nonzero-
1122 difference distribution (s.d. = 0.013), and it is one-sided, so on the full run 11 tasks clear it for the
1123 distributional score while 2 clear it for the mean and only the first number would conventionally be
1124 quoted. (ii) The seed is not a replicate. It re-draws which drugs are held out and it redefines the minority
1125 subpopulation by re-clustering, so a task recorded as dominant at one seed carries no information about
1126 robustness. (iii) When the drug is held fixed and the seed is resampled ten times, only 2 of the 10
1127 threshold-crossing cross-line tasks remain above threshold in at least 8 of 10 seeds, while control drugs
1128 that cross it at no seed in the recorded run cross it in 1 to 4 of 10 resamples. A control here is a drug that
1129 is dominant at no recorded seed; selecting controls by task identifier alone would let a drug that crosses
1130 at one seed and misses at another fall into both groups. The count is a count of noise excursions. The
1131 distribution is not, and it is what we report.

1132 Two caveats bound the projection. First, the regime boundary it transfers is fitted on HIR-Bench’s
1133 method-performance layer, which is available on a 144-instance grid because scoring nine retrieval meth-
1134 ods against up to 100 candidates per instance does not scale to the full grid. The boundary is therefore
1135 fitted on 72 grid cells, and we claim no more precision for it than that supports. Second, the projection’s
1136 own acceptance test fails. Its label derives from the same threshold we decline to trust, and it is heavily
1137 imbalanced (228 to 11), so a constant predictor scores 95.4% while the regime projection scores 61.1%,
1138 which is 34 points *worse* than saying nothing. The HIR-Bench boundary does not transfer to real data,
1139 and that holds whatever one makes of the threshold.

1140 Evaluation metric taxonomy and circularity audit

1141 We classify every evaluation metric into three classes. Class A: objective-aligned proxy metrics measure
1142 the same mathematical object the method optimizes or scores (energy-distance regret for energy-based
1143 retrieval, connectivity score for signature retrieval, reconstruction error for expression predictors, graph
1144 proximity for graph inverse design); they test objective fidelity, not independent utility. Class B: task-
1145 proximal metrics are biologically closer but still derived from the same upstream data (MoA recovery,
1146 target-family recovery, DEG overlap, pathway enrichment, minority-state coverage). Class C: oracle-
1147 independent utility metrics have external functional grounding (dose-response viability, longitudinal
1148 survival, post-treatment clone expansion, validated resistance emergence, functional readouts, combi-
1149 nation efficacy) and are required for therapeutic-utility claims.

1150 A field-level audit across six published method families, spanning signature retrieval^{8–10}, expres-
1151 sion predictors and foundation models^{2–6,14}, and graph/target inverse design^{7,15,16}, finds that every
1152 family’s *headline* metric falls in Class A and that none reports a Class-C readout as its primary evidence;
1153 population-level distributional metrics are themselves standard in recent perturbation-prediction bench-
1154 marks¹². The audit, its inclusion criteria, and the rule by which alignment risk is judged are given in full
1155 as Supplementary Table 2, so the claim can be checked rather than taken on assertion. It is an audit of
1156 *evaluation practice*, not of correctness, and the family is the unit: we do not score individual papers. We
1157 report the distributional score under all three classes. No Class-C readout exists within the single-cell
1158 datasets themselves; the Class-C analysis below therefore imports one, and we label it semi-real for that
1159 reason.

1160 Class-C analysis: an independent functional oracle and its scalar controls

1161 **Choosing the oracle.** A retrieval score is asked which candidate’s response distribution most resembles
1162 the query’s. It is not asked which candidate is most cytotoxic. Grading it on *absolute potency* therefore
1163 poses a question the method was never set, and it does so in a way that is not neutral: potency is largely
1164 driven by response magnitude, so a scalar that sorts candidates by magnitude is near-guaranteed to win,
1165 and a similarity retriever handed a weak query will correctly return other weak drugs and be scored as
1166 if it had failed. We therefore take the *semantically matched* oracle to be drug–drug functional similarity
1167 and report the absolute-potency comparison only as a confounder audit.

1168 **Construction.** SciPlex3 drugs were matched to GDSC records by normalized drug name (exact
1169 match, substring matching as a fallback), giving 35 matched compounds after removing one duplicate
1170 mapping (two SciPlex3 entries resolving to the same GDSC compound) and restricting to a *single* screen-
1171 ing platform, GDSC2. That platform restriction matters: GDSC1 and GDSC2 AUCs come from different
1172 assays and dose ranges and are not on a common scale, so pooling them would inject an assay-of-origin
1173 step into the oracle itself. For each matched compound we take its dose-response AUC profile across
1174 GDSC2 cell lines and define

$$\text{oracle}(q, c) = \text{Spearman}(\text{AUC}_{\bullet}(q), \text{AUC}_{\bullet}(c)),$$

1175 the rank correlation of the two compounds’ cross-cell-line sensitivity patterns. Three properties make
1176 this a usable Class-C oracle. It is *query-dependent*, so a query-independent scalar cannot game it. It is
1177 *out of context*: the three SciPlex3 lines (A549, K-562, MCF7) are *excluded* from the profile, leaving 966 cell

lines, so the oracle is never evaluated in the biological context the transcriptional data came from. And because a correlation is centered per compound, a drug’s overall potency *level* cannot drive it; only its selectivity *pattern* can, which is precisely the channel the absolute-potency comparison confounds.

Cell lines differ greatly in general drug sensitivity (mean AUC across all GDSC2 compounds ranges from 0.720 to 0.966), and that shared main effect inflates every drug–drug correlation: left uncorrected, 85% of drug pairs appear positively similar (median +0.178). We therefore subtract each cell line’s mean AUC, estimated across all 286 GDSC2 compounds rather than only the 35 matched ones, before correlating; the corrected oracle is symmetric about zero (52% positive, median +0.015). The correction was specified on this reasoning before the retrieval results were inspected. Both oracles are reported: they change every scorer’s absolute value but not the ordering that carries the conclusion. Pairs sharing fewer than 100 cell lines are dropped, leaving 591 usable drug pairs at a median of 909 shared lines.

Task. Per cell line, leave-one-drug-out: the held-out drug’s 10 μ M treated population is the query and the remaining matched drugs are the candidates (103 queries; 34–35 candidates per line). Each rule ranks the candidates and the ranking is compared with the oracle by Spearman correlation. The oracle is visible to no scorer.

The rankings and their controls. (i) *energy*, the distributional score. (ii) *mean cosine with control subtraction*, the correctly specified incumbent (treated mean minus matched control mean). (iii) *mean cosine without control subtraction*, how a mean baseline is often run in practice, which measures baseline cell identity rather than drug response. (iv) *magnitude matching*: rank by $-||d_c|| - ||d_q||$, preferring candidates whose response is about as large as the query’s. This is query-dependent yet compares no distributions whatsoever, and it is the null a distributional score must beat. (v) *magnitude only*: rank by $||d_c||$, which is query-independent and produces one ordering for every query; on a query-dependent oracle it is a falsifiable sanity check rather than a competitor. (vi) *potency matching*: rank by $-|AUC_c - AUC_q|$ in the query’s own line. This is a *diagnostic*, not a deployable recommender, since it requires the query drug’s own measured potency; it tests how far the functional oracle is re-encoding potency level, and we report that it does so substantially (median $\rho = +0.399$).

Reported statistics. Per-query Spearman correlations summarized by their median. Energy is additionally partialled on the magnitude-matching channel by rank-residualizing both the score and the oracle on it, which isolates the contribution attributable to the distribution rather than to magnitude. The 103 queries within a cell line share a candidate pool, so paired tests across them are anticonservative; we report such a *p*-value once, explicitly flagged, and treat the three cell lines as the honest unit, giving per-line medians throughout. We do not convert any rank correlation into a fraction of variance explained; a Spearman ρ is not one.

Confounder audit (absolute potency). The same rankings scored against absolute GDSC2 AUC are reported in Extended Data. There, energy is anti-correlated with potency ($\rho = -0.520$) and the query-independent magnitude scalar reaches +0.692. This follows arithmetically: response magnitude predicts potency ($\rho = -0.58$ to -0.71 by line), and the energy *distance* between query and candidate largely tracks the candidate’s own magnitude ($\rho = +0.791$), so ranking nearest-first returns the smallest-response candidates. We report this to expose the magnitude channel, and we draw no conclusion about therapeutic utility from it.

1218 The oracle-shape test: holding the data fixed and changing only the judge

1219 **Data.** Frangieh Perturb-CITE-seq measures RNA and 20 surface proteins (24 antibodies, 4 isotype con-
1220 trols removed) in the same cells. The two modalities are joined by cell barcode, not by row position: the
1221 RNA and protein matrices as distributed differ in row count, so a positional concatenation would pair
1222 each cell’s protein response with a different cell’s RNA, which produces a result that looks unremark-
1223 able and is pure noise. The join is asserted, giving 218,331 cells. Protein counts are CLR-normalized
1224 across antibodies per cell. Analyses are run *per immune condition* (Control, IFN γ , Co-culture; these live in
1225 *perturbation_2*, not *condition*), because pooling them confounds condition with perturbation. Knock-
1226 outs with at least 60 cells are used, subsampled to 250, seed fixed.

1227 **The two oracles.** Both are built from the *same* protein measurements of the *same* cells, and differ only
1228 in statistical form:

$$\text{oracle}_{\text{MEAN}}(q, c) = \cos(\bar{d}_q^{\text{prot}}, \bar{d}_c^{\text{prot}}), \quad \text{oracle}_{\text{DIST}}(q, c) = -\mathcal{E}(D_q^{\text{prot}}, D_c^{\text{prot}}),$$

1229 where $\bar{d}_k^{\text{prot}}$ is the mean protein response of knockout k against that condition’s control mean, D_k^{prot} is
1230 the set of *per-cell* protein responses of knockout k against the same control mean, and $\mathcal{E}$ is the energy
1231 distance. Centering both on the control is what makes the distributional oracle a *response* distance rather
1232 than a baseline-identity distance.

1233 **What is held fixed.** The two RNA rankings under test, energy and control-subtracted mean cosine,
1234 are computed once and scored against both oracles, so they are identical across the two columns by
1235 construction. No scorer sees protein data of any kind. Nothing but the oracle’s form changes.

1236 **The magnitude control, and why it is mandatory here.** Both distributional objects in this design
1237 are energy distances, and an energy distance largely tracks a candidate’s own response magnitude
1238 ($\rho = +0.791$; Extended Data Fig. 5). A magnitude-to-magnitude channel could therefore reproduce
1239 the reversal without any distribution being compared, and a paper that requires this control of others
1240 cannot omit it from its own sharpest claim. We therefore add a third RNA ranking, magnitude matching
1241 ($-\|d_c\| - \|d_q\|$), which is query-dependent and compares no distributions, and score it against both
1242 oracles. Reported statistics are per-query Spearman correlations summarized by their median, with the
1243 three immune conditions given separately as the honest unit.

1244 Resistance-associated exploratory analysis

1245 Response divergence per cell is correlated with curated resistance-associated program scores
1246 (AXL/mesenchymal, interferon-response, antigen-presentation) in the Frangieh data. This analysis is
1247 exploratory; no survival, timecourse, or functional readout is available, and no resistance-prediction
1248 claim is made.

1249 Natural heterogeneity: patient-derived glioblastoma

1250 **Data.** ZhaoSims2021: acute-slice culture and surgical biopsy from ten glioblastoma patients, treated
1251 with six compounds (Ana-12, Ispenisib, RO4929097, Tazemetostat, etoposide, panobinostat). Counts
1252 are normalized to 10^4 , log1p-transformed, and reduced to 2,000 highly variable genes. This is not a
1253 retrieval benchmark and is not used as one. Only one patient received more than two compounds, so a
1254 leave-one-drug-out retrieval would run on a single patient against five candidates; reporting Hit@1 from

1255 that would be an anecdote presented as a benchmark. It is used to test the two gates, and the paper's
1256 motivating premise, on heterogeneity that nobody constructed.

1257 **Compartment assignment, and why it is not an argmax.** The published annotation carries `celltype`
1258 = unknown for every cell, so compartments are assigned by canonical marker score: each cell receives
1259 a z-scored mean expression for each marker set, computed on the full log-normalized matrix before
1260 highly-variable-gene selection so the labelling never depends on a gene set chosen for downstream vari-
1261 ance. A first version of this procedure took the bare argmax over five marker sets, which *forces* every cell
1262 into some compartment including cells that express no markers at all. It produced two compartments
1263 that fail their own validation: cells labelled T cells (22% of the dataset) had a mean CD3D/CD3E/CD2
1264 expression of 0.080, *lower* than their malignant-marker score of 0.141, and cells labelled endothelial ex-
1265 pressed myeloid markers more strongly than endothelial ones. They were neither; they were unidenti-
1266 fiable cells swept into whichever class the argmax had left over. Building a natural-heterogeneity result
1267 on a fabricated T-cell compartment would be worse than having no natural dataset.

1268 A cell is therefore assigned to a compartment only if it scores highest for that compartment, clears
1269 an absolute expression floor on that compartment's key markers (≥ 0.25 mean log-normalized expres-
1270 sion), and beats the runner-up by a z-score margin (≥ 0.25). Everything else is labelled unassigned and
1271 *excluded*, not reassigned; 39.9% of cells are dropped this way. The natural-tissue result is insensitive to
1272 these two constants. Over floor = margin $\in \{0.10, 0.25, 0.40, 0.50\}$, which drops between 29% and 58%
1273 of cells and passes compartment validation at every setting, the Gate-2 ceiling stays 0.915–0.923, the
1274 best unsupervised method 0.77–0.80, the gap 0.11–0.13, and the Gate-1 induced cosine 0.54–0.59 (analy-
1275 sis/natural/zhao_threshold_sensitivity.py). The 0.25 setting is neither the most nor the least favourable.
1276 Only the three compartments that pass validation are used: malignant glioma cells (own-marker expres-
1277 sion 1.02 versus 0.07 and 0.12 for the others), tumour-associated myeloid cells (1.53 versus 0.12, 0.17)
1278 and oligodendrocytes (1.78 versus 0.41, 0.11). The loader raises rather than writing labels if any called
1279 compartment fails to express its own key markers most strongly, and the validation table ships with the
1280 results. Labels are never shown to any clustering method or to any retrieval score; they exist only to
1281 evaluate whether those methods recover them.

1282 **Gate 1 (natural).** For each (patient, drug) with at least 50 cells in both the malignant and the myeloid
1283 compartment, $d_k = \text{mean}(\text{treated cells of compartment } k) - \text{mean}(\text{that patient's control cells of compartment } k)$,
1284 and the reported quantity is $\cos(d_{\text{malignant}}, d_{\text{myeloid}})$. Controls are matched *within patient and within com-*
1285 *partment*, so the cosine cannot be inflated by a shared patient or compartment effect.

1286 **Gate 2 (natural), and the comparison our first attempt got wrong.** Gate 2 asks whether the subpop-
1287 ulations *inside a candidate population* can be told apart, which is a *drug-response* partition. Our first natural
1288 test instead separated malignant from myeloid *control* cells, a *cell-type* partition between two lineages
1289 that differ in thousands of genes. It is a far easier problem, it returned a ceiling of 0.964, and comparing
1290 that with the constructed mixture's 0.692 compared two different questions while holding the protocol
1291 fixed. That number is withdrawn.

1292 The reported analysis (analysis/natural/gate2_drug_response.py) poses the same question in both
1293 settings. A split is a pair of drugs: cells treated with drug A against cells treated with drug B, *within one*
1294 *compartment and within one patient*, at least 50 cells per drug, capped at 150 per source, 5 seeds. This gives
1295 36 drug-pair splits over 4 patients and 2 compartments. The constructed comparator is run at the same

granularity, one drug against one drug in K562 at 10 μ M (24 HDAC-versus-JAK drug pairs, subsampled with a fixed seed from the 513 available and reported as subsampled). The class-pooled split the paper previously used is retained as a third arm, because a pooled class carries the extra within-class variance of several distinct compounds and is therefore harder for a reason that has nothing to do with the mixture being constructed. Quoting it against the natural arm would inflate the contrast.

The representation, and a bias we had stated backwards. Everything else follows the constructed-mixture protocol: a common 30-component PCA for the unsupervised methods, best-permutation accuracy as the shared unit with label matching given to each clusterer for free, and a supervised ceiling taken as the best of logistic regression, random forest and k -nearest-neighbours under 5-fold cross-validation. We report the ceiling under two representations. The *in-fold* ceiling re-fits PCA inside each training fold. This is the leak-free choice, but it is not the representation the clusterers were given; because it lowers the ceiling it *shrinks* the gap, which is the direction that supports the conclusion the gap is used to draw. The *matched* ceiling uses the same transductive PCA the clusterers receive, which leaks no label information because PCA never sees labels, and it is the apples-to-apples quantity; it is what we report. *The reported gap is the median of the paired per-split differences*, not the difference of the two marginal medians; a median is not linear and the two disagree (for the natural arm, +0.117 against +0.147), and the paired statistic is the correct and the smaller one.

Uncertainty on the gap (analysis/natural/gate2_uncertainty.py). Three things make the raw number fragile and each is addressed. (i) The gap is a difference of two *maxima*, best of three probes minus best of four clusterers, and a maximum over methods is upward-biased. We therefore also compute every fixed (probe, clusterer) pair, in which no selection occurs; HDBSCAN is excluded from the summary because it assigns every cell to noise and scores exactly 0.500, so a gap measured against it reflects a clusterer that did nothing rather than information that is unreachable. (ii) The confidence interval is a cluster bootstrap over patients, not over splits. Thirty of the 36 natural splits come from one patient and reuse the same six drugs, so a split-level bootstrap would treat dependent observations as independent and return an interval several times too narrow, which is precisely the pseudo-replication this paper criticises elsewhere. With only four clusters the bootstrap is coarse and is known to under-cover, so the interval is reported as a floor on the uncertainty rather than a faithful estimate of it. (iii) The dominant patient is dropped entirely and the analysis repeated on the remaining three. The resulting values are given in Fig. 5d, with two that appear only here: the maximisation moves the natural number by only -0.005 , and the natural-minus-constructed difference is $+0.109$ [$+0.052$, $+0.129$], with none of 5,000 bootstrap draws at or below zero.

The premise. For every pair of drugs within a patient, we compute the cosine between their population *mean* signatures and the cosine between their *malignant-compartment* responses, and report the Spearman correlation between the two across drug pairs. If the mean were a sufficient statistic these would be the same number. Because the mean signature contains the malignant cells, 43% of those called, we also report the *disjoint* form, in which the malignant-compartment response is ranked by the myeloid-compartment response instead; the two compartments share no cells and each is referred to its own compartment-matched control (analysis/natural/zhao_premise_disjoint.py). *No pass/fail threshold is applied.* An earlier version of the analysis script declared the premise supported if the correlation fell below 0.9, and the observed value was 0.878. That threshold was ours, it was applied after seeing the

number, and clearing it by 0.02 licenses nothing; the verdict has been removed from the code and the correlation is reported and interpreted in the text.

Re-measuring the three conditions on Tahoe-100M

Why this dataset. The three conditions are measured elsewhere in this paper on cell-line mixtures we assembled and on a tumour cohort in which one patient supplies 15 of 18 drug pairs. Tahoe-100M³⁹ is used for neither biology nor relevance but for sample size, and because its heterogeneity is *unconstructed*: within one cell line the cell-to-cell variation is intrinsic state, not two populations combined by us. Plate 3 is a complete grid, 50 cell lines $\times$ 93 compounds at a single concentration each, with a DMSO_TF vehicle arm in every line, 4,158,278 cells and a median of 648 cells per (line, compound). We use the released preprocessed matrix as it ships, 2,304 highly variable genes in log1p space, and refit nothing.

Partitions, and why two. Gates 1 and 3 need a subpopulation split inside a treated population. The first is the cell-cycle call shipped with the data (G1 versus G2M; S is dropped as intermediate). The second is fitted independently: k -means with $k = 2$ on each line's *control* cells only, with treated cells assigned to the nearest control centroid, so the partition is a property of untreated state and is fixed before any treated cell is seen and cannot be tuned to the response it is used to measure. Both are reported everywhere; neither is selected after the fact. Gate 2 uses neither, and instead uses drug identity, which is external to expression: recovering a marker-derived phase call from expression would be partly circular, and circularity is the subject of this paper. That makes the Gate 2 test here the same construct as the K562 benchmark and the tumour test, so the three are comparable.

Statistics. Each subpopulation's response is its mean minus the mean of the *same-subpopulation* control cells, so a baseline difference between subpopulations cannot enter the response (the error recorded as R28 and R30). Gate 1 is the cosine between the two subpopulation responses. Gate 3 correlates, across all within-line compound pairs, the cosine similarity of the majority responses with the cosine similarity of the minority responses; the two sides share *no cells*, which is the form we also apply to the tumour statistic above. Gate 2 draws a balanced sample of up to 300 cells per compound for 20 random compound pairs per line, scores a supervised ceiling as the out-of-fold accuracy of a five-fold cross-validated logistic probe and an unsupervised arm as the best of k -means in gene space, k -means and Gaussian mixtures in 50-dimensional PCA space, and Leiden with its resolution reduced until it returns two communities. Every method receives best-permutation label matching for free, as elsewhere in this paper. A condition enters Gates 1 and 3 only with at least 50 cells in each of the four arms and a pair enters Gate 2 only with at least 100 cells per arm; every exclusion is counted and the resulting bias is reported in Results rather than assumed away. Seed 0 throughout.

What we did not do. We did not run retrieval on this dataset, and we report no Hit@ k , no regret and no Class-A, Class-B or Class-C comparison from it. It enters this paper only as a larger measurement of the three conditions.

Identifiability phase diagram

The phase diagram sweeps a separation scale and a cell budget on the K562 HDAC-versus-JAK mixture, whose labels are known by construction. Separation is applied by pushing each source along the vector from its centroid to the global centroid. Those centroids are computed once, on the *full* source populations, so the manipulation is independent of the sample being evaluated. This is not a detail: computing

the centroids on the sampled cells and then shifting those same cells fits the manipulation on the data it is measured on, amplifies whatever separation each small draw happens to contain, and produces the indefensible result that the adjusted Rand index *falls* as the cell budget rises. Clustering does not become harder with more cells.

Statistical analysis

Paired comparisons use the Wilcoxon signed-rank test; group comparisons use Mann–Whitney U; monotone associations use Spearman correlation. Multiple comparisons across strata use Benjamini–Hochberg false-discovery-rate control⁴⁰, and we report achieved statistical power and the sample size required for 80% power where effects are small. Effect sizes are reported alongside *p*-values throughout, and statistically significant but negligible effects are labeled as such. Where an effect is reported as negligible, the operating range of the metric is given so that the effect size can be read against it.

Data availability

All datasets analysed in this study are publicly available.

Three of the four single-cell datasets were obtained in harmonized form from scPerturb¹⁰, Zenodo record 13350497 (DOI [10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4), and we name the exact file for each so the analysis can be reproduced from the same bytes rather than from a re-derivation of the primary data. SciPlex3: file SrivatsanTrapnell2020_sciplex3.h5ad; primary data in Gene Expression Omnibus under accession GSE139944³⁰. Frangieh melanoma Perturb-CITE-seq: files FrangiehIzar2021_RNA.h5ad and FrangiehIzar2021_protein.h5ad (the protein modality is used by the surface-protein oracle); original data³¹. ZhaoSims2021 patient-derived glioblastoma (the natural-heterogeneity benchmark): file ZhaoSims2021.h5ad; original data⁴¹. CD34+ HSPC: Gene Expression Omnibus, accession GSE306429.

Tahoe-100M (the unconstructed-heterogeneity check on the three conditions): released by Vevo Therapeutics and the Arc Institute under CC0 and distributed through the Arc Virtual Cell Atlas and Hugging Face (tahoebio/Tahoe-100M)³⁹. This study uses plate 3 only, in the released preprocessed form (4,158,278 cells × 2,304 highly variable genes, log1p), together with the released obs_metadata and sample_metadata tables. No number in this paper is derived from the remaining thirteen plates.

GDSC dose-response (the Class-C functional oracle): Genomics of Drug Sensitivity in Cancer, GDSC2 fitted dose-response, release 8.5 (file GDSC2_fitted_dose_response_270ct23.xlsx), available from <https://www.cancerrxgene.org> and the Sanger bulk-download mirror. The specific release was recovered and verified: the retained join table (drug_match_table.csv) reproduces the AUC and ln IC₅₀ of this release exactly across all checked (compound, cell line) pairs, and the Class-C statistics were re-derived from it to confirm they are unchanged. The join table is released with the code so the drug matching can be audited independently.

LINCS L1000 / PDGrapher benchmark: a precomputed benchmark table ships with the code (pdgrapher_closed_loop_benchmark.parquet). No reported result in this paper derives from it, so it is not data underlying the findings; we name it only because it is present in the release, and we do not have a retained accession for it.

The processed tensors used by the experiments, together with the scripts that build them from the primary data, are described in the repository. Source data for every figure panel are provided as the experiment output tables in the code repository.

1417 Code availability

1418 All code, experiment scripts, and the evaluation-metric taxonomy are available at [https://github.com/](https://github.com/Boom5426/EvalShift)
1419 [Boom5426/EvalShift](https://github.com/Boom5426/EvalShift) ([**archived DOI on Zenodo at acceptance**]). The release includes a test suite
1420 and a numerical consistency check that re-derives every reported constant from the cached experiment
1421 outputs; the latter re-reads the cached experiment tables and diffs them against the recorded constants,
1422 and is a consistency check rather than an end-to-end re-execution. A single driver script re-executes the
1423 pipeline from the processed tensors. The Tahoe-100M re-measurement of the three conditions is separate
1424 and self-contained (analysis/tahoe_pilot/): one script computes the gate statistics, one recomputes
1425 Gate 3 on disjoint cell sets under both partitions, one re-scores Gate 2 across the full clusterer panel,
1426 and one derives every Tahoe number quoted in this paper from the per-condition and per-pair tables it
1427 writes, so each can be rechecked without rerunning the others.

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1430 Author contributions

1431 [**Author contributions, CRediT taxonomy: conceptualization, methodology, software, formal analy-**
1432 **sis, investigation, data curation, writing (original draft), writing (review and editing), visualization,**
1433 **supervision, funding acquisition.**]

1434 Competing interests

1435 [**Competing interests statement. If none: “The authors declare no competing interests.”**]

1436 Additional information

1437 Supplementary Information is available for this paper: Extended Data Figs. 1–6, Supplementary Notes
1438 1–3 and Supplementary Tables 1–4.

1439 Correspondence and requests for materials should be addressed to [**Name**].

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